The essential and the annihilator part: material for section 2.2 of *Metric and topological homology*, with the accompanying insertions in chapters 1 and 3

Book-ready form of the companion note

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This file is the companion note in the form in which it drops into [B]. It has four parts. Part I holds the prose paragraphs that belong in chapter 1, each headed by where it goes. Part II is the new section 2.2, "The essential and the annihilator part", in six subsections; at the head of each subsection the statements of [B] that are to be interleaved are listed by label, with the point of insertion. Part III holds the material for chapter 3. Part IV is the merge manifest and the list of preamble additions.

Throughout, $\mathrm{sep}_X(x) = \mathrm{dist}(x, X_{ess})$ is the *separation* of a vector from the essential part and $\mathrm{sepd}(X) = \sup\{\mathrm{sep}_X(x) : x \in B_{X_{ann}}\}$ its *separation defect*; replacing annihilator vectors by nets annihilated only asymptotically gives a second defect $\mathrm{asepd}(X)$ and the chain $\mathrm{sepd}(X) \le \mathrm{sepd}(X^*) \le \mathrm{asepd}(X)$. Two constants of the algebra arise by evaluating the defect on the free module of rank one and on its dual,

$$\mathrm{sepd}(A_+) = \frac{1}{1 + c_{\mathrm{ri}}(A)}, \qquad \mathrm{sepd}(A_+^*) = \frac{1}{1 + c_{\mathrm{rai}}(A)},$$

and the three homological properties then obey one estimate with two constants: a $\langle$ metrically / C-topologically $\rangle$ projective P, injective J or flat F satisfies $\mathrm{sepd}(P) \le \langle\, \mathrm{sepd}(A_+) \,/\, C\,\mathrm{sepd}(A_+) \,\rangle$ and $\mathrm{sepd}(J), \mathrm{sepd}(F) \le \langle\, \mathrm{sepd}(A_+^*) \,/\, C\,\mathrm{sepd}(A_+^*) \,\rangle$. Since sepd $= 1$ on a non-zero annihilator module, these lines return the algebra clauses of the annihilator characterisations collected in subsection 2.2.6 — a right identity in the projective case, a bounded right approximate identity in the other two.

1 For chapter 1

The paragraphs below are prose: chapter 1 of [B] carries no numbered statement anywhere, and these do not change that. Each is headed by its destination.

1.1 For subsection SubSectionBanachSpaces, in the paragraph that already carries $E^*/F^\perp \underset{\mathbf{Ban}_1}{\cong} F^*$ and $(E/F)^* \underset{\mathbf{Ban}_1}{\cong} F^\perp$

For a non-empty subset M of a Banach space E and a vector $x \in E$ we write $\mathrm{dist}(x, M) = \inf\{\|x - y\| : y \in M\}$. The function $\mathrm{dist}(\cdot, M) : E \to \mathbb{R}_+$ is 1-Lipschitz, and it is positively

homogeneous whenever M is a linear subspace. If M is a closed subspace, the Hahn-Banach theorem gives the dual description

$$\operatorname{dist}(x, M) = \max\{|f(x)| : f \in B_{M^\perp}\},$$

the maximum being attained; combined with the isometric isomorphism $i : (E/M)^* \to M^\perp$ above, this says that

$$\operatorname{dist}(x, M) = \|x + M\|_{E/M} = \max\{|f(x)| : f \in B_{(E/M)^*}\}$$

We shall use three further classical facts about weak topologies. By the Banach-Alaoglu theorem the closed unit ball of a dual space is weak* compact. By Goldstine's theorem $\iota_E(B_E)$ is weak* dense in $B_{E^{**}}$. By Mazur's theorem the weak closure of a convex subset of E coincides with its norm closure.

1.2 For subsection `SubSectionBanachAlgebrasAndTheirModules`, after the paragraph introducing approximate identities

It is convenient to name the optimal bounds. By $\langle\ c_{\mathrm{ri}}(A)\ /\ c_{\mathrm{li}}(A)\ \rangle$ we denote the infimum of the norms of the $\langle$ right / left $\rangle$ identities of A, and by $\langle\ c_{\mathrm{rai}}(A)\ /\ c_{\mathrm{lai}}(A)\ \rangle$ the infimum of the bounds of the $\langle$ right / left $\rangle$ approximate identities of A; each of the four is $+\infty$ if there is nothing to take the infimum over, and each of the four is 0 for the zero algebra, whose only element is a two-sided identity and whose only net is a two-sided approximate identity. For $A \neq \{0\}$ all four constants lie in $[1, +\infty]$. If A is commutative, a left approximate identity is a right one and conversely, so $c_{\mathrm{lai}}(A) = c_{\mathrm{rai}}(A)$; it is not known whether $c_{\mathrm{lai}}(A) = c_{\mathrm{rai}}(A)$ whenever both are finite. If A is unital then

$$c_{\mathrm{ri}}(A) = c_{\mathrm{li}}(A) = c_{\mathrm{rai}}(A) = c_{\mathrm{lai}}(A) = \|e_A\|,$$

the zero algebra included, with $e_A = 0$ and all four constants 0. Finally, when $\langle\ c_{\mathrm{lai}}(A)\ /\ c_{\mathrm{rai}}(A)\ \rangle$ is finite it is attained: some $\langle$ left / right $\rangle$approximate identity is bounded by $\langle\ c_{\mathrm{lai}}(A)\ /\ c_{\mathrm{rai}}(A)\ \rangle$ itself. This is theorem 2.11 below.

1.3 After introduction of essential and faithful modules add the following facts

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Suppose A has a $\langle$ left / right $\rangle$ approximate identity $(e_\nu)_{\nu \in N}$. For any $\langle$ left / right $\rangle$ Banach A-module X and any $x \in X_{ess}$ the net $\langle\ (e_\nu \cdot x)_{\nu \in N}\ /\ (x \cdot e_\nu)_{\nu \in N}\ \rangle$ converges to x. If Y is another $\langle$ left / right $\rangle$ Banach A-module and $\phi : X \to Y$ is an A-mprphism, then $\phi(X_{ess}) \subset Y_{ess}$ and $\phi(X_{ann}) \subset Y_{ann}$.

1.4 Move this to the new subsection 2.2 before proposition 2.4 of this note as a numbered definition

An *annihilator net* in a $\langle$ left / right $\rangle$ Banach A-module X is a net $(x_\nu)_{\nu \in N}$ in X such that $\langle\ (a \cdot x_\nu)_{\nu \in N}\ /\ (x_\nu \cdot a)_{\nu \in N}\ \rangle$ is norm null for every $a \in A$.

1.5 For subsection `SubSectionBanachAlgebrasAndTheirModules`, in the paragraph that already records $(X^*)^{\perp S} = \mathrm{cl}_X(SX)^\perp$

Setting $S = A$ in the identity above and reading it on the two sides at once: for a left Banach A-module X the dual right module X^* has $(X^*)_{ann} = (X_{ess})^\perp$. The two subspaces $(X^*)_{ess}$ and $(X_{ann})^\perp$ are in general different, though the first is always contained in the second.

2 The essential and the annihilator part (the new section 2.2)

The subsection headings below are those of the target structure. A new section 2.2 is inserted between the present sections 2.1 and 2.2 of [B]; the present section 2.2 becomes 2.3 and the present 2.3 becomes 2.4, save for the present subsection 2.2.1, which becomes subsection 2.2.6 below.

2.1 Separation

Nothing of [B] is interleaved here.

Definition 2.1. Let X be a ⟨ left / right ⟩ Banach A-module. The *separation* of a vector $x \in X$ is

$$\mathrm{sep}_X(x) := \mathrm{dist}(x, X_{ess}),$$

and the *separation defect* of X is

$$\mathrm{sepd}(X) := \sup\{\mathrm{sep}_X(x) : x \in B_{X_{ann}}\}.$$

For $x \in X$ we write $\mathrm{sng}(x) := \inf\{\|a \cdot x\|/\|a\| : a \in A \setminus \{0\}\}$ for the *singularity* of x, so for $x \in X_{ann}$ we have $\mathrm{sng}(x) = 0$.

The essential part is a closed linear subspace, so sep_X is 1-Lipschitz and positively homogeneous; we use both properties below without further comment. In particular $\mathrm{sepd}(X) = \sup\{\mathrm{sep}_X(x)/\|x\| : x \in X_{ann} \setminus \{0\}\}$ as soon as $X_{ann} \neq \{0\}$, and $\mathrm{sepd}(X) = \mathrm{sep}_X(0) = 0$ otherwise. The defect measures the directed gap from the annihilator part to the essential part. The degenerate case is not idle: modules with $X_{ann} = \{0\}$ but $X_{ess} \neq X$ occur constantly and for them $\mathrm{sepd}(X) = 0$ while the module is far from essential.

Lemma 2.2. *Suppose A has a ⟨ left / right ⟩ approximate identity bounded by C. Then in every ⟨ left / right ⟩ Banach A-module X, for any $x \in X_{ann}$ holds*

$$\mathrm{sep}_X(x) \geq \frac{\|x\|}{1+C} \qquad (x \in X_{ann}).$$

Note: one can put ⟨ $C = c_{\mathrm{lai}}(A)$ / $C = c_{\mathrm{rai}}(A)$ ⟩.

Proof. We write the left-hand case, the right-hand one is a trivial adaptation. Let $(e_\nu)_{\nu \in N}$ be a left approximate identity of A with $\sup_\nu \|e_\nu\| \leq C$. Dispose first of $C = 0$: then $A = \{0\}$, $X_{ess} = 0$. So the assertion reads holds with equality. Assume from now on $C > 0$.

Now let $x \in X_{ann}$ and $y \in X_{ess}$. Since $e_\nu \cdot x = 0$, then

$$\|e_\nu \cdot y\| = \|e_\nu \cdot (y - x)\| \leq \|e_\nu\|\|x - y\| \leq K\|x - y\|,$$

and letting ν run through N turns the left-hand side into $\|g\|$. So $\|x - y\| \geq \|y\|/C$, while the triangle inequality gives $\|x - y\| \geq \|x\| - \|y\|$. If $\|y\| \geq C\|x\|/(1 + C)$ the first estimate gives $\|x - y\| \geq \|x\|/(1 + C)$; if $\|y\| < K\|x\|/(1 + C)$ the second gives the same. Taking the infimum over $y \in X_{ess}$ yields $\mathrm{sep}_X(x) \geq \|x\|/(1 + C)$. By taking supremum over all upper norm bounds of all left approximate identities one can set ⟨ $C = c_{\mathrm{lai}}(A)$ / $C = c_{\mathrm{rai}}(A)$ ⟩. □

Proposition 2.3. *For every ⟨ left / right ⟩ Banach A-module X one has* $\mathrm{sepd}(X) \leq \mathrm{sepd}(X^*)$.

Proof. If $X_{ann} = \{0\}$ then $\operatorname{sepd}(X) = 0$ and there is nothing to prove. Assume $X_{ann} \neq \{0\}$. By the Hahn-Banach description of the distance to a closed subspace and the identity $(X^*)_{ann} = (X_{ess})^\perp$, we get

$$\operatorname{sep}_X(x) = \operatorname{dist}(x, X_{ess}) = \max\{|f(x)| : f \in B_{(X_{ess})^\perp}\} = \max\{|f(x)| : f \in B_{(X^*)_{ann}}\}$$

for all $x \in X$. Therefore,

$$\operatorname{sepd}(X) = \sup\{\operatorname{sep}_X(x) : x \in X_{ann}\} = \sup\{|f(x)| : x \in X_{ann}, f \in B_{(X^*)_{ann}}\}$$

On the other hand $(X_{ess})^* \subset (X_{ann})^\perp$. So for $f \in (X^*)_{ann} \setminus \{0\}$ holds

$$\operatorname{sep}_{X^*}(f) = \operatorname{dist}\big(f, (X^*)_{ess}\big) \geq \operatorname{dist}\big(f, (X_{ann})^\perp\big) = \|f + (X_{ann})^\perp\|_{X/(X_{ann})^\perp} = \big\|f|_{X_{ann}}\big\|,$$

the last equality being the isometric identification of $X^*/Y^\perp$ with Y^* by restriction, applied to the closed subspace $Y = X_{ann}$. Taking the supremum over $f \in B_{(X^*)_{ann}}$ and comparing with the previous calculation gives $\operatorname{sepd}(X^*) \geq \sup\big\{\|f|_{X_{ann}}\| : f \in B_{(X^*)_{ann}}\big\} = \operatorname{sepd}(X)$. □

The gap between $(X^*)_{ess}$ and $(X_{ann})^\perp$ is exactly the slack in proposition 2.3. Since $(X^*)_{ann} = (X_{ess})^\perp$, the defect of the dual is $\operatorname{sepd}(X^*) = \sup\{\operatorname{sep}_{X^*}(f) : f \in (X_{ess})^\perp, \|f\| \leq 1\}$, and the next proposition computes each of these numbers out of nets in X that are annihilated only asymptotically. That is what makes it computable.

Proposition 2.4. *Let X be a $\langle$ left / right $\rangle$ A-module and $f \in (X_{ess})^\perp$. Then*

(i) $\operatorname{sep}_{X^*}(f) = \max\{|q(f)| : q \in B_{((X^*)_{ess})^\perp}\}$;

(ii) $\operatorname{sep}_{X^*}(f) = \sup\big\{\limsup_\nu |f(x_\nu)| : (x_\nu)_{\nu\in N}$ *an annihilator net in* $B_X\big\}$;

Proof. *(i)* This is a direct conssequence of the formula for distance to $(X^*)_{ess}$ in terms of functionals vanishing on $(X^*)_{ess}$.

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(ii) Writing $L_a : X \to X$, $x \mapsto a \cdot x$, a bounded operator with $L_a^* g = g \cdot a$, we have $E = \operatorname{cl}_{X^*}(\operatorname{span}\{L_a^* g\})$, and since $\langle L_a^{**} q, g\rangle = \langle q, L_a^* g\rangle$ the condition $q|_E = 0$ holds if and only if $L_a^{**} q = 0$ for every $a \in A$.

Nets give at most the distance. Let $(x_\nu)_{\nu\in N}$ be an annihilator net in B_X and put $L = \limsup_\nu |f(x_\nu)|$. Refining twice — once so that $|f(x_\nu)| \to L$, once by Banach-Alaoglu so that $\iota_X(x_\nu) \to q$ weak* for some $q \in B_{X^{**}}$ — and using that a subnet of an annihilator net is one, we get $\langle L_a^{**} q, g\rangle = \lim_\nu g(a \cdot x_\nu) = 0$ for all $a \in A$ and $g \in X^*$, because $\|a \cdot x_\nu\| \to 0$. So q is admissible and $|q(f)| = L$.

The distance is attained along a net. Let q attain the maximum; after multiplying by a unimodular scalar we may assume $q(f) = \operatorname{sep}_{X^*}(f)$. By Goldstine's theorem there is a net (y_μ) in B_X with $\iota_X(y_\mu) \to q$ weak*; then $f(y_\mu) \to q(f)$ and, for all $a \in A$ and $g \in X^*$, $g(a \cdot y_\mu) = (g \cdot a)(y_\mu) \to q(g \cdot a) = 0$ because $g \cdot a \in E$. That is, $a \cdot y_\mu \to 0$ weakly for every $a \in A$. Fix a finite $F \subset A$ and $\epsilon > 0$; the set $\{\big(f(x), (a \cdot x)_{a\in F}\big) : x \in B_X\} \subset \mathbb{C} \times X^F$ is convex, being the image of B_X under a linear map, and $\big(q(f), 0\big)$ lies in its weak closure, hence by Mazur's theorem in its norm closure; so there is $x \in B_X$ with $|f(x) - q(f)| < \epsilon$ and $\|a \cdot x\| < \epsilon$ for every $a \in F$. Directing the pairs (F, ϵ) by $(F, \epsilon) \preceq (F', \epsilon')$ when $F \subset F'$ and $\epsilon' \leq \epsilon$, and choosing such a vector at each index, produces an annihilator net in B_X along which $|f(\cdot)|$ tends to $|q(f)| = \operatorname{sep}_{X^*}(f)$. □

Definition 2.5. The *asymptotic separation* of an annihilator net $(x_\nu)_{\nu\in N}$ in a left A-module X is $\operatorname{asep}_X\big((x_\nu)_{\nu\in N}\big) := \limsup_\nu \operatorname{sep}_X(x_\nu)$, and the *asymptotic separation defect* of X is

$$\operatorname{asepd}(X) := \sup\big\{\operatorname{asep}_X\big((x_\nu)_{\nu\in N}\big) : (x_\nu)_{\nu\in N} \text{ an annihilator net in } B_X\big\} \in [0,1].$$

Proposition 2.6. *For every left Banach A-module X one has* $\operatorname{sepd}(X) \le \operatorname{sepd}(X^*) \le \operatorname{asepd}(X)$.

Proof. The first inequality is proposition 2.3. For the second, let $f \in (X_{ess})^\perp$ with $\|f\| \le 1$; if 0 is the only such f there is nothing to prove. For $x \in X$ and $g \in X_{ess}$ one has $f(x) = f(x-g)$, so $|f(x)| \le \|x-g\|$, and taking the infimum over g gives $|f(x)| \le \operatorname{sep}_X(x)$. Hence $\limsup_\nu |f(x_\nu)| \le \limsup_\nu \operatorname{sep}_X(x_\nu) \le \operatorname{asepd}(X)$ for every annihilator net in B_X, and proposition 2.4 computes $\operatorname{sep}_{X^*}(f)$ as the supremum of the left-hand side over such nets. Taking the supremum over $f \in B_{(X^*)_{ann}}$ gives $\operatorname{sepd}(X^*) \le \operatorname{asepd}(X)$. □

The three terms ask the same question with increasing generosity: exact annihilator vectors, annihilator directions in the dual, asymptotically annihilated nets. Both inequalities can be strict, and the next theorem says when they all collapse.

Theorem 2.7. *Let X be a left Banach A-module such that every annihilator net (x_ν) in B_X satisfies* $\operatorname{dist}(x_\nu, X_{ann}) \to 0$*; this holds for every module when A is unital. Then* $\operatorname{sepd}(X) = \operatorname{sepd}(X^*) = \operatorname{asepd}(X)$.

Proof. By proposition 2.6 it suffices to prove $\operatorname{asepd}(X) \le \operatorname{sepd}(X)$. Let $(x_\nu)_{\nu\in N}$ be an annihilator net in B_X and $\epsilon > 0$. By hypothesis there is ν_0 such that for $\nu \ge \nu_0$ one may choose $y_\nu \in X_{ann}$ with $\|x_\nu - y_\nu\| < \epsilon$, and then $\|y_\nu\| \le 1+\epsilon$; homogeneity of sep_X and the definition of sepd give $\operatorname{sep}_X(y_\nu) \le (1+\epsilon)\operatorname{sepd}(X)$, trivially so when $y_\nu = 0$. The Lipschitz property then gives $\operatorname{sep}_X(x_\nu) \le (1+\epsilon)\operatorname{sepd}(X) + \epsilon$ for $\nu \ge \nu_0$. Take the limit superior in ν, the supremum over annihilator nets in B_X, and let $\epsilon \to 0$.

Finally, let A be unital and (x_ν) any annihilator net in B_X. Put $y_\nu := x_\nu - e_A \cdot x_\nu$; then $a \cdot y_\nu = a \cdot x_\nu - (ae_A)\cdot x_\nu = 0$ for every $a \in A$, so $y_\nu \in X_{ann}$, while $\|x_\nu - y_\nu\| = \|e_A \cdot x_\nu\| \to 0$. □

Both defects are computed on $\bigoplus_1$- and $\bigoplus_\infty$-sums by two facts: the annihilator part of a sum is the sum of the annihilator parts, and the distance to a coordinatewise closed subspace is the sum, respectively the supremum, of the coordinate distances. The essential part is the sum of the essential parts only in the first case. The lower bounds all come from one observation, which we isolate because it is used again for free, cofree and injective modules.

Lemma 2.8. *Let $\pi : X \to Y$ and $\sigma : Y \to X$ be contractive A-morphisms with $\pi\sigma = 1_Y$. Then $\sigma(Y_{ann}) \subset X_{ann}$, $\pi(X_{ess}) \subset Y_{ess}$ and $\operatorname{sep}_X(\sigma(y)) \ge \operatorname{sep}_Y(y)$ for every $y \in Y$; consequently* $\operatorname{sepd}(X) \ge \operatorname{sepd}(Y)$ *and* $\operatorname{asepd}(X) \ge \operatorname{asepd}(Y)$.

Proof. For $y \in Y_{ann}$ one has $a \cdot \sigma(y) = \sigma(a\cdot y) = 0$, and $\pi(a\cdot x) = a \cdot \pi(x) \in \operatorname{span}(AY)$ while Y_{ess} is closed. Hence for $y \in Y$ and $z \in X_{ess}$,

$$\|\sigma(y) - z\| \ge \|\pi\sigma(y) - \pi(z)\| = \|y - \pi(z)\| \ge \operatorname{sep}_Y(y),$$

and taking the infimum over z gives $\operatorname{sep}_X(\sigma(y)) \ge \operatorname{sep}_Y(y)$. Since also $\|\sigma(y)\| \le \|y\|$, restricting to $y \in Y_{ann}$ and forming the quotient of the two gives $\operatorname{sepd}(X) \ge \operatorname{sepd}(Y)$. Finally, if (y_ν) is an annihilator net in B_Y then so is $(\sigma(y_\nu))$ in B_X, and $\limsup_\nu \operatorname{sep}_X(\sigma(y_\nu)) \ge \limsup_\nu \operatorname{sep}_Y(y_\nu)$; hence $\operatorname{asepd}(X) \ge \operatorname{asepd}(Y)$. □

Proposition 2.9. *Let $(X_\lambda)_{\lambda\in\Lambda}$ be left Banach A-modules with $\Lambda \neq \varnothing$, and let $X = \bigoplus_p\{X_\lambda : \lambda \in \Lambda\}$ with the coordinatewise action, for $p = 1$ or $p = \infty$. Then $X_{ann} = \bigoplus_p\{(X_\lambda)_{ann} : \lambda \in \Lambda\}$ for both p, and*

(i) $\operatorname{sepd}\big(\bigoplus_1\{X_\lambda : \lambda \in \Lambda\}\big) = \sup_\lambda \operatorname{sepd}(X_\lambda)$;

(ii) $\operatorname{sepd}\big(\bigoplus_\infty\{X_\lambda : \lambda \in \Lambda\}\big) \geq \sup_\lambda \operatorname{sepd}(X_\lambda)$, *with equality when Λ is finite;*

(iii) $\operatorname{asepd}\big(\bigoplus_p\{X_\lambda : \lambda \in \Lambda\}\big) \geq \sup_\lambda \operatorname{asepd}(X_\lambda)$ *for $p = 1$ and for $p = \infty$, with equality when Λ is finite;*

(iv) for $p = 1$, $X_{ess} = \bigoplus_1\{(X_\lambda)_{ess} : \lambda \in \Lambda\}$ and $\operatorname{sep}_X(x) = \sum_\lambda \operatorname{sep}_{X_\lambda}(x_\lambda)$ for every $x \in X$.

Proof. The action is contractive for both p, and $a \cdot x = 0$ for every a if and only if $a \cdot x_\lambda = 0$ for every a and every λ, which is the description of X_{ann}. Each X_μ is a 1-retract of X via the coordinate injection and projection, so lemma 2.8 gives all three lower bounds.

(iv) The inclusion $X_{ess} \subset \bigoplus_1\{(X_\lambda)_{ess}\}$ holds because $a \cdot x$ lies in the right-hand side, a closed subspace. Conversely, given y in the right-hand side and $\epsilon > 0$, choose a finite $S \subset \Lambda$ with n elements such that $\sum_{\lambda\notin S}\|y_\lambda\| < \epsilon/2$ and, for $\lambda \in S$, $z_\lambda \in \operatorname{span}(AX_\lambda)$ with $\|y_\lambda - z_\lambda\| < \epsilon/(2n)$; the vector with these coordinates on S and 0 elsewhere lies in $\operatorname{span}(AX)$ and is within ϵ of y. Next, if $M_\lambda \subset X_\lambda$ are closed subspaces and $M = \bigoplus_1\{M_\lambda\}$, then $\operatorname{dist}(x, M) = \sum_\lambda \operatorname{dist}(x_\lambda, M_\lambda)$: the inequality $\geq$ is coordinatewise, and for $\leq$ one picks $m_\lambda \in M_\lambda$ with $\|x_\lambda - m_\lambda\| \leq \operatorname{dist}(x_\lambda, M_\lambda) + \epsilon_\lambda$ for a summable family (ϵ_λ), the resulting m being summable because $\|m_\lambda\| \leq 2\|x_\lambda\| + \epsilon_\lambda$. Together the two give the formula for sep_X.

(i) For $x \in X_{ann}$ each x_λ lies in $(X_\lambda)_{ann}$, so $\operatorname{sep}_{X_\lambda}(x_\lambda) \leq \operatorname{sepd}(X_\lambda)\|x_\lambda\|$ and *(iv)* gives $\operatorname{sep}_X(x) \leq \big(\sup_\mu \operatorname{sepd}(X_\mu)\big)\|x\|$.

(ii) For $p = \infty$ the inclusion $X_{ess} \subset \bigoplus_\infty\{(X_\lambda)_{ess}\}$ holds by the argument of *(iv)*, but the reverse can fail, since $\operatorname{cl}_X(AX)$ need not reach every bounded family of essential vectors; and distances to a $\bigoplus_\infty$-sum of closed subspaces dominate the supremum of the coordinate distances. Distances to a larger set being smaller, this already gives $\operatorname{sep}_X(x) \geq \sup_\lambda \operatorname{sep}_{X_\lambda}(x_\lambda)$. When Λ is finite there is no tail to discard, so the argument of *(iv)* gives $X_{ess} = \bigoplus_\infty\{(X_\lambda)_{ess}\}$, while a near-optimal choice in each of the finitely many coordinates gives $\operatorname{dist}(x, M) = \max_\lambda \operatorname{dist}(x_\lambda, M_\lambda)$; hence $\operatorname{sep}_X(x) = \max_\lambda \operatorname{sep}_{X_\lambda}(x_\lambda)$, which for $x \in X_{ann}$ is the equality in *(ii)*.

(iii) Let Λ be finite and (x^ν) an annihilator net in B_X. Pass to a subnet along which $\operatorname{sep}_X(x^\nu)$ converges to its limit superior and $\|x^\nu_\lambda\| \to t_\lambda$ for each of the finitely many λ; then $\sum_\lambda t_\lambda \leq 1$ for $p = 1$ and $t_\lambda \leq 1$ for $p = \infty$. If $t_\lambda = 0$ then $\operatorname{sep}_{X_\lambda}(x^\nu_\lambda) \leq \|x^\nu_\lambda\| \to 0$. If $t_\lambda > 0$ then eventually $\|x^\nu_\lambda\| \geq t_\lambda/2 > 0$ and the normalised vectors $x^\nu_\lambda/\|x^\nu_\lambda\|$ form an annihilator net in B_{X_λ}, whence $\limsup_\nu \operatorname{sep}_{X_\lambda}(x^\nu_\lambda) \leq t_\lambda \operatorname{asepd}(X_\lambda)$. Now *(iv)* for $p = 1$ and *(ii)* for $p = \infty$, together with the fact that the limit superior of a finite sum, respectively of a finite maximum, is at most the sum, respectively the maximum, of the limits superior, give $\limsup_\nu \operatorname{sep}_X(x^\nu) \leq \sup_\lambda \operatorname{asepd}(X_\lambda)$. Finiteness enters exactly here: it is what forces the masses t_λ to account for the whole net. □

2.2 The separation of the unitization

Nothing of [B] is interleaved here.

Evaluating the separation defect on the free module of rank one, A_+, and on its dual produces two constants of the algebra. They differ exactly as an identity differs from an approximate one, and the dual is the more generous of the two for a structural reason: the dual of A_+ always has an annihilator direction, namely the augmentation $\varepsilon : A_+ \to \mathbb{C}$, $a \oplus_1 \lambda \mapsto \lambda$, whereas A_+ itself has one only when A has a right identity.

Lemma 2.10. *Regard A_+ as a left A-module and A_+^* as the dual right A-module. Then $(A_+)_{ess} = A$, so that $\operatorname{sep}_{A_+}(v) = |\varepsilon(v)|$ for every $v \in A_+$; and a vector $y = b \oplus_1 \lambda$ lies in $(A_+)_{ann}$ if and only if either $\lambda = 0$ and $Ab = \{0\}$, or $\lambda \neq 0$ and $u := -b/\lambda$ is a right identity of A, in which case $y = \lambda(e_{A_+} - u)$. Consequently*

$$\operatorname{sepd}(A_+) = \frac{1}{1 + c_{\mathrm{ri}}(A)},$$

which is positive if and only if A has a right identity. Moreover $(A_+^*)_{ann} = A^\perp = \mathbb{C}\varepsilon$ with $\|\varepsilon\| = 1$, so that $\operatorname{sepd}(A_+^*) = \operatorname{sep}_{A_+^*}(\varepsilon)$: a single distance in a single direction.

Proof. In $A_+ = A \bigoplus_1 \mathbb{C}$ the left action of A is $a \cdot (b \oplus_1 \lambda) = (ab + \lambda a) \oplus_1 0$, so $A \cdot A_+ \subset A$; conversely $a = a \cdot e_{A_+}$, so $\operatorname{span}(A \cdot A_+) = A$, which is closed. Hence $(A_+)_{ess} = A$ and, for $v = b \oplus_1 \lambda$, $\operatorname{dist}(v, A) = \inf_{c \in A}(\|b - c\| + |\lambda|) = |\varepsilon(v)|$.

The same formula shows that $y = b \oplus_1 \lambda$ is annihilated by A if and only if $ab + \lambda a = 0$ for every $a \in A$. If $\lambda = 0$ this says $Ab = \{0\}$, and then $y \in A = (A_+)_{ess}$, so $\operatorname{sep}_{A_+}(y) = 0$. If $\lambda \neq 0$, put $u := -b/\lambda \in A$; the condition becomes $au = a$ for every a, so u is a right identity and $y = \lambda(e_{A_+} - u)$; conversely every vector of that form is annihilated. For such a y, $\operatorname{sep}_{A_+}(y) = |\lambda|$ while $\|y\| = |\lambda|(1 + \|u\|)$, so the ratio $\operatorname{sep}_{A_+}(y)/\|y\|$ is $1/(1 + \|u\|)$, the annihilator vectors with $\lambda = 0$ contributing 0. Hence $\operatorname{sepd}(A_+)$ is the supremum of $1/(1+\|u\|)$ over the right identities u of A, which is $1/(1+c_{\mathrm{ri}}(A))$, the supremum over the empty set being 0 and matching the convention $c_{\mathrm{ri}}(A) = +\infty$.

Finally, $(A_+^*)_{ann} = \big((A_+)_{ess}\big)^\perp = A^\perp$; a functional on $A \bigoplus_1 \mathbb{C}$ vanishing on A is determined by its value at e_{A_+}, so $A^\perp = \mathbb{C}\varepsilon$, and $\|\varepsilon\| = 1$. The defect of A_+^* is therefore the separation of the single unit vector ε. □

The dual constant is computed by approximate identities instead, and there the infimum is attained. No hypothesis on A is needed: all three clauses hold for the zero algebra, where $A_+ = \mathbb{C}$ has $(A_+)_{ess} = \{0\}$ and $(A_+^*)_{ess} = \{0\}$, so that $\operatorname{sepd}(A_+^*) = \|\varepsilon\| = 1$, matching $1/(1+c_{\mathrm{rai}}(\{0\})) = 1$, and the zero net attains the bound.

Theorem 2.11. *(i)* $\operatorname{sep}_{X^*}(f) \geq \|f\|/(1 + c_{\mathrm{rai}}(A))$ *for every left A-module X and every $f \in (X_{ess})^\perp$; equivalently* $\operatorname{sepd}(X^*) \geq \operatorname{sepd}(A_+^*)$ *whenever X is not essential;*

(ii) $\operatorname{sepd}(A_+^*) = \dfrac{1}{1 + c_{\mathrm{rai}}(A)}$*;*

(iii) $c_{\mathrm{rai}}(A)$, when finite, is attained by a right approximate identity.

Proof. *(i)* The dual X^* is a right A-module with $(X^*)_{ann} = (X_{ess})^\perp$, so lemma 2.2 in its right-hand form, applied to X^*, gives the displayed bound; if A has no right approximate identity then $c_{\mathrm{rai}}(A) = +\infty$ and the bound is vacuous. For the reformulation, $\operatorname{sepd}(X^*)$ is the supremum of $\operatorname{sep}_{X^*}(f)/\|f\|$ over $f \in (X^*)_{ann} \setminus \{0\}$, and that set is non-empty exactly when $X_{ess} \neq X$, by the Hahn-Banach theorem; the bound then reads $\operatorname{sepd}(X^*) \geq 1/(1 + c_{\mathrm{rai}}(A))$, which is $\operatorname{sepd}(A_+^*)$ by *(ii)*, proved below and independently of *(i)*.

(ii) By lemma 2.10 we have $\operatorname{sepd}(A_+^*) = \operatorname{sep}_{A_+^*}(\varepsilon)$, so proposition 2.4 at $X = A_+$ and $f = \varepsilon$ reads

$$\operatorname{sepd}(A_+^*) = \sup\Big\{\limsup_\nu |\lambda_\nu| : \|b_\nu\| + |\lambda_\nu| \leq 1,\ \lim_\nu \|ab_\nu + \lambda_\nu a\| = 0 \text{ for all } a \in A\Big\},$$

the supremum being over nets $v_\nu = b_\nu \oplus_1 \lambda_\nu$, since $\varepsilon(v_\nu) = \lambda_\nu$ and $a \cdot v_\nu = (ab_\nu + \lambda_\nu a) \oplus_1 0$. Call such a net admissible.

For the lower bound, suppose A has a right approximate identity and let $\epsilon > 0$. Choose one, $(e_\nu)_{\nu\in N}$, with $\sup_\nu \|e_\nu\| \le c_{\mathrm{rai}}(A)+\epsilon$, and put $v_\nu := (e_{A_+} - e_\nu)/(1+\|e_\nu\|)$. These are unit vectors, hence admissible; $a\cdot v_\nu = (a - ae_\nu)/(1+\|e_\nu\|) \to 0$; and $\lambda_\nu = 1/(1+\|e_\nu\|) \ge 1/(1+c_{\mathrm{rai}}(A)+\epsilon)$. Let $\epsilon \to 0$.

For the upper bound, let (v_ν) be admissible with $\limsup_\nu |\lambda_\nu| = d > 0$; pass to a subnet along which $|\lambda_\nu| \to d$, so that eventually $\lambda_\nu \neq 0$, and set $u_\nu := -b_\nu/\lambda_\nu \in A$. Then

$$\|a - au_\nu\| = \frac{\|a\cdot v_\nu\|}{|\lambda_\nu|} \longrightarrow 0 \quad (a\in A), \qquad \|u_\nu\| = \frac{\|b_\nu\|}{|\lambda_\nu|} \le \frac{1-|\lambda_\nu|}{|\lambda_\nu|} \longrightarrow \frac{1-d}{d},$$

so (u_ν) is a right approximate identity and, for every $\epsilon > 0$, its tail on which $\|u_\nu\| \le (1-d)/d + \epsilon$ is one as well; hence $c_{\mathrm{rai}}(A) \le (1-d)/d$, that is $d \le 1/(1+c_{\mathrm{rai}}(A))$. The same computation shows that $d > 0$ is impossible when A has no right approximate identity, so $\mathrm{sepd}(A_+^*) = 0$ there, again the value of $1/(1+c_{\mathrm{rai}}(A))$.

(iii) Assume $c_{\mathrm{rai}}(A) < +\infty$ and put $\delta := \mathrm{sepd}(A_+^*) > 0$. With $E = (A_+^*)_{ess}$, proposition 2.4 says that the distance $\delta = \mathrm{dist}(\varepsilon, E)$ is attained at some $q \in B_{A_+^{**}}$ with $q|_E = 0$; multiplying q by a unimodular scalar we may assume $q(\varepsilon) = \delta$. Since $A_+ = A \bigoplus_1 \mathbb{C}$ we have $A_+^* \underset{\mathbf{Ban}_1}{\cong} A^* \bigoplus_\infty \mathbb{C}$ and $A_+^{**} \underset{\mathbf{Ban}_1}{\cong} A^{**} \bigoplus_1 \mathbb{C}$, where $g \in A_+^*$ is written $g = (h, c)$ with $h = g|_A$ and $c = g(e_{A_+})$, so that $g = h + c\varepsilon$ and $\langle (m,t),(h,c)\rangle = m(h) + tc$. In these coordinates $q = (m,\delta)$ with $\|m\| + \delta \le 1$, that is $\|m\| \le 1-\delta$. The right action reads

$$(g\cdot a)(b \oplus_1 \lambda) = g\big((ab + \lambda a) \oplus_1 0\big) = (h\cdot a)(b) + \lambda\, h(a),$$

that is $g\cdot a = \big(h\cdot a,\ h(a)\big)$, which does not involve c; since E is the closed span of the $g\cdot a$, the condition $q|_E = 0$ says $m(h\cdot a) + \delta h(a) = 0$ for all $h \in A^*$ and $a \in A$. Setting $p := -m/\delta \in A^{**}$ we get

$$p(h\cdot a) = h(a) \quad (h\in A^*,\ a\in A), \qquad \|p\| = \frac{\|m\|}{\delta} \le \frac{1-\delta}{\delta} = c_{\mathrm{rai}}(A).$$

By Goldstine's theorem applied to the ball of radius $\|p\|$ there is a net (x_μ) in A with $\|x_\mu\| \le \|p\|$ and $\iota_A(x_\mu) \to p$ weak*; then $h(ax_\mu) = (h\cdot a)(x_\mu) \to p(h\cdot a) = h(a)$, so $ax_\mu \to a$ weakly for every $a \in A$. A Mazur argument on the convex set $\{(ax - a)_{a\in F} : x \in A,\ \|x\| \le \|p\|\}$, followed by directing the pairs (F,ϵ) as in the proof of proposition 2.4, produces a right approximate identity bounded by $\|p\| \le c_{\mathrm{rai}}(A)$; since $c_{\mathrm{rai}}(A)$ is the infimum of such bounds, this one attains it. □

Lemma 2.12. *Let $A \neq \{0\}$, let $a \in A$ with $\|a\| = 1$, let $\varphi \in A_+^*$ satisfy $\|\varphi\| = 1$ and $\varphi(a) = 1$, and let $0 < \gamma \le 1$. Then $\|\varepsilon - \gamma\,\varphi\cdot a\| \le \max\{\gamma, 1-\gamma\}$. In particular $\mathrm{sepd}(A_+^*) \le \frac{1}{2}$ for every $A \neq \{0\}$.*

Proof. A vector a of norm one exists because $A \neq \{0\}$, and such a φ exists because on the line $\mathbb{C}a$ the functional $ta \mapsto t$ has norm one and the Hahn-Banach theorem extends it to A_+. For $b \oplus_1 \lambda \in A_+$ one has $a\cdot(b\oplus_1\lambda) = (ab + \lambda a) \oplus_1 0$, so, using $\varphi(a) = 1$,

$$(\varepsilon - \gamma\,\varphi\cdot a)(b\oplus_1\lambda) = \lambda - \gamma\varphi(ab) - \gamma\lambda = \lambda(1-\gamma) - \gamma\varphi(ab),$$

whose modulus is at most $(1-\gamma)|\lambda| + \gamma\|b\| \le \max\{\gamma,1-\gamma\}\big(|\lambda| + \|b\|\big)$. Since $\varphi\cdot a \in (A_+^*)_{ess}$, lemma 2.10 gives $\mathrm{sepd}(A_+^*) = \mathrm{sep}_{A_+^*}(\varepsilon) \le \|\varepsilon - \gamma\,\varphi\cdot a\| \le \max\{\gamma,1-\gamma\}$, which at $\gamma = \frac{1}{2}$ is $\frac{1}{2}$. □

Corollary 2.13. *Let $A \neq \{0\}$. Then $\mathrm{sepd}(A_+^*) = \frac{1}{2}$ if and only if A has a contractive right approximate identity; $\mathrm{sepd}(A_+^*) = 0$ if and only if A has no right approximate identity, if and only if $\varepsilon \in (A_+^*)_{ess}$; and $\mathrm{sepd}(A_+^*) = 1/(1+\|e_A\|)$ for unital A.*

Proof. By theorem 2.11(*ii*) all three clauses are statements about $c_{\mathrm{rai}}(A)$, which for $A \neq \{0\}$ satisfies $c_{\mathrm{rai}}(A) \geq 1$ by subsection `SubSectionBanachAlgebrasAndTheirModules` — the same ceiling $\operatorname{sepd}(A_+^*) \leq \frac{1}{2}$ that lemma 2.12 gives independently. So $\operatorname{sepd}(A_+^*) = \frac{1}{2}$ if and only if $c_{\mathrm{rai}}(A) = 1$, which holds if A has a contractive right approximate identity and, conversely, forces one to exist by theorem 2.11(*iii*). Next, $\operatorname{sepd}(A_+^*) = 0$ if and only if $c_{\mathrm{rai}}(A) = +\infty$, that is, if and only if A has no right approximate identity; and $\operatorname{sepd}(A_+^*) = \operatorname{dist}\big(\varepsilon, (A_+^*)_{ess}\big)$ vanishes exactly when ε lies in that closed subspace. The unital clause is $c_{\mathrm{rai}}(A) = \|e_A\|$, recorded in subsection `SubSectionBanachAlgebrasAndTheirModules`.

The hypothesis $A \neq \{0\}$ is not decoration, and this is the only place in this section where it bites: for $A = \{0\}$ the conventions give $c_{\mathrm{rai}}(A) = 0$ while $\operatorname{sepd}(A_+^*) = 1$, so the first clause would fail, the zero net being a contractive right approximate identity. □

The dual constant runs over the whole of $(0, \frac{1}{2}]$. Let $A_t := (\mathbb{C}, t|\cdot|)$, the field $\mathbb{C}$ with the norm $\|z\| = t|z|$; this is a Banach algebra exactly for $t \geq 1$, since $\|zw\| = t|zw|$ while $\|z\|\|w\| = t^2|zw|$. Its identity is 1, of norm t, so the unital clause of corollary 2.13 gives $\operatorname{sepd}\big((A_t)_+^*\big) = 1/(1+t)$, and as t runs over $[1, +\infty)$ this runs over $(0, \frac{1}{2}]$. The identity map $A_t \to A_s$ is a topological isomorphism of Banach algebras for all $s, t \geq 1$, so $\operatorname{sepd}(A_+^*)$ is an isometric and not a topological invariant of A.

It depends on the side as well. Let $R = \left\{ \left(\begin{smallmatrix} a & b \\ 0 & 0 \end{smallmatrix}\right) \right\} \subset M_2(\mathbb{C})$, a subalgebra because $\left(\begin{smallmatrix} a & b \\ 0 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} a' & b' \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} aa' & ab' \\ 0 & 0 \end{smallmatrix}\right)$. The matrix unit E_{11} lies in R, has norm one and satisfies $E_{11}x = x$ for every $x \in R$, so $c_{\mathrm{li}}(R) = c_{\mathrm{lai}}(R) = 1$ and the mirror constant, computed from left approximate identities, is $\frac{1}{2}$. On the other side, $E_{12} \in R$ satisfies $E_{12}x = 0$ for every $x \in R$, so no net (e_ν) in R can have $E_{12}e_\nu \to E_{12} \neq 0$: the algebra R has no right approximate identity, $c_{\mathrm{rai}}(R) = +\infty$, and $\operatorname{sepd}\big(R_+^*\big) = 0$.

One reading of proposition `MetTopFlatCycModCharac` of [B] is worth recording here, since it is the first place in [B] where the dual constant occurs unnamed. For a closed left ideal I of A_+ the quantity $\sup_\nu \|e_{A_+} - e_\nu\|$ appearing there equals $1 + \sup_\nu \|e_\nu\|$, because $e_\nu \in I \subset A$; its infimum over the right bounded approximate identities of I is therefore $1 + c_{\mathrm{rai}}(I) = 1/\operatorname{sepd}\big(I_+^*\big)$, and that proposition is the computation of this constant along the coset of the identity. Proposition 2.4 and theorem 2.11 give it directly.

Both constants are constants of A_+ alone: free modules of higher rank add nothing to either. For sepd this is proposition 2.9(*i*), a free module being an ℓ_1-sum of copies of A_+; the asymptotic defect needs an argument.

Proposition 2.14. $\operatorname{asepd}\big(A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda)\big) = \operatorname{sepd}(A_+^*)$ *for every* $\Lambda \neq \varnothing$*, a value at most* $\frac{1}{2}$ *when* $A \neq \{0\}$*: free modules have the minimal asymptotic separation defect.*

Proof. First, $\operatorname{asepd}(A_+) = \operatorname{sepd}(A_+^*)$. Indeed $\operatorname{sep}_{A_+}(v) = |\varepsilon(v)|$ by lemma 2.10, so an annihilator net (v_ν) in B_{A_+} has $\operatorname{asep}_{A_+}\big((v_\nu)\big) = \limsup_\nu |\varepsilon(v_\nu)|$; taking the supremum over such nets, definition 2.5 and proposition 2.4 at (A_+, ε) give $\operatorname{asepd}(A_+) = \operatorname{sep}_{A_+^*}(\varepsilon) = \operatorname{sepd}(A_+^*)$.

Write $P = A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$. The canonical map carrying $u = \sum_\lambda u_\lambda \otimes \delta_\lambda$ to (u_λ) is an isometric isomorphism of left A-modules onto $\bigoplus_1 \{A_+ : \lambda \in \Lambda\}$, so by proposition 2.9(*iv*) and lemma 2.10 the essential part of P is $A \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$ and, writing $u_\lambda = b_\lambda \oplus_1 \beta_\lambda$,

$$\operatorname{sep}_P(u) = \sum_\lambda \operatorname{sep}_{A_+}(u_\lambda) = \sum_\lambda |\beta_\lambda|.$$

A coordinate copy of A_+ is a 1-retract of P, so lemma 2.8 gives $\operatorname{asepd}(P) \geq \operatorname{asepd}(A_+)$.

For the upper bound let $(u^\nu)_{\nu\in N}$ be an annihilator net in B_P. For each ν and λ choose a unimodular c_λ^ν with $c_\lambda^\nu\beta_\lambda^\nu = |\beta_\lambda^\nu|$ and put $w_\nu := \sum_\lambda c_\lambda^\nu u_\lambda^\nu \in A_+$, a series converging absolutely because $\sum_\lambda \|u_\lambda^\nu\| = \|u^\nu\| \le 1$. Then $\|w_\nu\| \le 1$ and $\|a\cdot w_\nu\| \le \sum_\lambda \|a\cdot u_\lambda^\nu\| = \|a\cdot u^\nu\| \to 0$, so (w_ν) is an annihilator net in B_{A_+}; and, ε being linear and continuous, $\varepsilon(w_\nu) = \sum_\lambda |\beta_\lambda^\nu| = \operatorname{sep}_P(u^\nu)$. Hence $\limsup_\nu \operatorname{sep}_P(u^\nu) = \limsup_\nu |\varepsilon(w_\nu)|$, which proposition 2.4 at (A_+,ε) bounds by $\operatorname{sepd}(A_+^*)$. Taking the supremum over annihilator nets in B_P gives the stated equality, and the bound $\frac{1}{2}$ for $A \neq \{0\}$ is lemma 2.12. □

The dual of the free module of rank Λ is the cofree module of rank Λ, and its separation defect is the second constant, not the first. The upper bound comes from the predual: that is the only use the chain of proposition 2.6 gets outside subsection 2.1.

Proposition 2.15. *For every left A-module X that is not essential,*

$$\operatorname{sepd}(A_+^*) \;\le\; \operatorname{sepd}(X^*) \;\le\; \operatorname{asepd}(X),$$

so $\operatorname{sepd}(X^*) = \operatorname{sepd}(A_+^*)$ *as soon as X attains the minimal asymptotic separation defect. In particular, for every $\Lambda \neq \varnothing$ the cofree right A-module satisfies*

$$\operatorname{sepd}\big(\mathcal{B}(A_+, \ell_\infty(\Lambda))\big) = \operatorname{sepd}(A_+^*) = \frac{1}{1+c_{\mathrm{rai}}(A)}.$$

Proof. If X is not essential then $(X_{ess})^\perp \neq \{0\}$ by the Hahn-Banach theorem, so theorem 2.11(i) gives $\operatorname{sepd}(X^*) \ge \operatorname{sepd}(A_+^*)$, while proposition 2.6 gives $\operatorname{sepd}(X^*) \le \operatorname{asepd}(X)$; if $\operatorname{asepd}(X) = \operatorname{sepd}(A_+^*)$ the outer terms coincide.

Write $F_\Lambda = A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$. Since $\ell_\infty(\Lambda) \underset{\mathbf{Ban}_1}{\cong} \ell_1(\Lambda)^*$, the law of adjoint associativity gives isometric isomorphisms of right A-modules

$$\mathcal{B}\big(A_+, \ell_\infty(\Lambda)\big) \underset{\mathbf{mod}_1\text{-}A}{\cong} \mathcal{B}\big(A_+, \mathcal{B}(\ell_1(\Lambda),\mathbb{C})\big) \underset{\mathbf{mod}_1\text{-}A}{\cong} \mathcal{B}\big(A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda), \mathbb{C}\big) = F_\Lambda^*.$$

By proposition 2.9(iv) and lemma 2.10 the essential part of F_Λ is $A \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$, a proper closed subspace since A is proper in A_+ and $\Lambda \neq \varnothing$; so F_Λ is not essential, and proposition 2.14 gives $\operatorname{asepd}(F_\Lambda) = \operatorname{sepd}(A_+^*)$. The two bounds therefore meet at $X = F_\Lambda$, and the value is $1/(1+c_{\mathrm{rai}}(A))$ by theorem 2.11(ii). □

So the free modules see the identities of A and the cofree ones its approximate identities: $\operatorname{sepd}(F_\Lambda) = \operatorname{sepd}(A_+) = 1/(1+c_{\mathrm{ri}}(A))$ by proposition 2.9(i) and lemma 2.10, against $\operatorname{sepd}(F_\Lambda^*) = 1/(1+c_{\mathrm{rai}}(A))$. Corollary 2.24 will recover the second value in one line and for all metrically injective modules at once.

2.3 The annihilator part of a projective module

Interleaved from [B], in this order, before the material below: proposition `MetTopRelProjModCompIdealPartCompl`, corollary `MetTopRelProjModPartCompl`, proposition `MetTopProjModEssQuotIsProjBanSp`.

Those three say that the essential part of a projective module is complemented and that the quotient by it is a projective Banach space. What follows measures how far the annihilator part stands out of the essential one, and it does so through a single lifting problem: the augmentation $\varepsilon : A_+ \to \mathbb{C}$. The retract characterisation `MetCTopProjModViaCanonicMorph` is not needed anywhere in this subsection.

Lemma 2.16. *Let P be a $\langle$ metrically / C-topologically $\rangle$ projective left A-module. Then for every $x \in P$ there is an A-morphism $\psi : P \to A_+$ with $\|\psi\| \leq \langle\, 1 \,/\, C \,\rangle$ and $\varepsilon(\psi(x)) = \mathrm{sep}_P(x)$.*

Proof. Regard $\mathbb{C}$ as a left A-module with the zero action. By the Hahn-Banach description of $\mathrm{sep}_P(x) = \mathrm{dist}(x, P_{ess})$ there is $g \in (P_{ess})^\perp$ with $\|g\| \leq 1$ and $g(x) = \mathrm{sep}_P(x)$; take $g = 0$ if $x \in P_{ess}$. Then $g : P \to \mathbb{C}$ is an A-morphism, since $a \cdot y \in P_{ess}$ gives $g(a \cdot y) = 0 = a \cdot g(y)$. For the same two reasons ε is an A-morphism, $\varepsilon(a \cdot v) = 0 = a \cdot \varepsilon(v)$; it has norm one and carries B_{A_+} onto $B_{\mathbb{C}}$, since $\varepsilon(0 \oplus_1 \mu) = \mu$ with $\|0 \oplus_1 \mu\| = |\mu|$, so it is strictly coisometric and in particular strictly 1-topologically surjective. By definition `MetCTopProjMod`, with $c = 1$ in the topological bracket, projectivity of P lifts g along ε: there is an A-morphism $\psi : P \to A_+$ with $\varepsilon\psi = g$ and $\|\psi\| \leq \langle\, \|g\| \,/\, C\|g\| \,\rangle \leq \langle\, 1 \,/\, C \,\rangle$. Finally $\varepsilon(\psi(x)) = g(x) = \mathrm{sep}_P(x)$. □

Theorem 2.17. *Let P be a $\langle$ metrically / C-topologically $\rangle$ projective left A-module. Then*

$$\mathrm{sep}_P(x) \leq \langle\, \mathrm{sepd}(A_+) \,/\, C\, \mathrm{sepd}(A_+) \,\rangle \|x\| \qquad (x \in P_{ann}),$$

that is, $\mathrm{sepd}(P) \leq \langle\, \mathrm{sepd}(A_+) \,/\, C\, \mathrm{sepd}(A_+) \,\rangle$.

Proof. Fix $x \in P_{ann}$ and let ψ be as in lemma 2.16. The vector $\psi(x)$ lies in $(A_+)_{ann}$, because $a \cdot \psi(x) = \psi(a \cdot x) = 0$ for every $a \in A$, so lemma 2.10 and the definition of sepd give

$$\mathrm{sep}_P(x) = \varepsilon(\psi(x)) = \mathrm{sep}_{A_+}\big(\psi(x)\big) \leq \mathrm{sepd}(A_+)\|\psi(x)\| \leq \langle\, \mathrm{sepd}(A_+) \,/\, C\, \mathrm{sepd}(A_+) \,\rangle \|x\|.$$

Dividing by $\|x\|$ and taking the supremum over $x \in P_{ann} \setminus \{0\}$ gives the statement about $\mathrm{sepd}(P)$. □

One lifting problem suffices, and it is the one dual to the extension problem that governs injectivity: projectivity meets the annihilator module $\mathbb{C} \underset{A\text{-}\mathbf{mod}_1}{\cong} A_+/A$ as a *quotient* of A_+ and lifts along the augmentation, while injectivity and flatness meet its dual $\mathbb{C}\varepsilon = A^\perp$ as a *subobject* of A_+^* and extend along the inclusion or tensor with it — see propositions 2.21 and 2.25(ii). That the projective constant is the smaller of the two — proposition 2.3 at $X = A_+$ — is nothing but the difference between splitting the augmentation sequence exactly and splitting it approximately.

Corollary 2.18. *Let $P \neq \{0\}$ be an annihilator A-module which is $\langle$ metrically / C-topologically $\rangle$ projective. Then $\langle\, A = \{0\} \,/\, c_{\mathrm{ri}}(A) \leq C - 1 \,\rangle$.*

Proof. Since P is a non-zero annihilator module, $P_{ess} = \{0\}$ and $P_{ann} = P$, so $\mathrm{sep}_P(x) = \|x\|$ for every $x \in P$ and $\mathrm{sepd}(P) = 1$. Theorem 2.17 therefore gives $\mathrm{sepd}(A_+) \geq \langle\, 1 \,/\, 1/C \,\rangle$, and lemma 2.10 turns this into a statement about $c_{\mathrm{ri}}(A)$ through $\mathrm{sepd}(A_+) = 1/(1 + c_{\mathrm{ri}}(A))$. In the topological bracket it reads $c_{\mathrm{ri}}(A) \leq C - 1$. In the metric bracket it reads $c_{\mathrm{ri}}(A) \leq 0$; but $c_{\mathrm{ri}}(A) \geq 1$ for $A \neq \{0\}$ by subsection `SubSectionBanachAlgebrasAndTheirModules`, so only $A = \{0\}$ is compatible with it. □

Corollary 2.18 is stated as a necessary condition only. Its converse is the implication $(ii) \Rightarrow (i)$ of proposition `MetTopProjOfAnnihModCharac`, into which the corollary is dissolved in subsection 2.6; it is recorded here because it is what theorem 2.17 contributes, and it is not cited anywhere else in this subsection.

Corollary 2.19. *Let A have no right identity. Then* $\mathrm{sepd}(A_+) = 0$*, so in every topologically projective A-module P one has $P_{ann} \subset P_{ess}$, and no non-zero annihilator module is projective, metrically or topologically.*

Proof. If A has no right identity then $c_{\mathrm{ri}}(A) = +\infty$, so $\operatorname{sepd}(A_+) = 0$ by lemma 2.10. Let P be C-topologically projective for some C and let $x \in P_{ann}$; theorem 2.17 gives $\operatorname{sep}_P(x) \le C \operatorname{sepd}(A_+)\|x\| = 0$, and P_{ess} is closed, so $x \in P_{ess}$.

For the last assertion let $P \neq \{0\}$ be an annihilator module. Then $P_{ess} = \{0\}$ and $P_{ann} = P$, so $\operatorname{sepd}(P) = 1$, and theorem 2.17 forces $\operatorname{sepd}(A_+) \ge \langle\, 1 \,/\, 1/C \,\rangle > 0$ in either bracket, contradicting $\operatorname{sepd}(A_+) = 0$. □

On the asymptotic layer the answer changes, and it is the dual constant that appears.

Proposition 2.20. *Let $A \neq \{0\}$ and let P be a $\langle$ metrically / C-topologically $\rangle$ projective left A-module. If P is essential, then* $\operatorname{asepd}(P) = 0$*; otherwise*

$$\langle\, \operatorname{asepd}(P) = \operatorname{sepd}(A_+^*) \;/\; \operatorname{sepd}(A_+^*) \le \operatorname{asepd}(P) \le C \operatorname{sepd}(A_+^*) \,\rangle.$$

Proof. If P is essential then sep_P vanishes identically and $\operatorname{asepd}(P) = 0$.

Let P be non-essential, let $(x_\nu)_{\nu\in N}$ be an annihilator net in B_P and write $M := \langle\, 1 \,/\, C \,\rangle$, a number at least 1. For each ν let ψ_ν be the morphism supplied by lemma 2.16 at x_ν and put $v_\nu := \psi_\nu(x_\nu)$. Then $\|v_\nu\| \le M$ and $\|a \cdot v_\nu\| = \|\psi_\nu(a \cdot x_\nu)\| \le M\|a \cdot x_\nu\| \to 0$, so (v_ν/M) is an annihilator net in B_{A_+}, while $\varepsilon(v_\nu) = \operatorname{sep}_P(x_\nu)$. Proposition 2.4 at (A_+, ε) therefore gives $\limsup_\nu \operatorname{sep}_P(x_\nu)/M \le \operatorname{sep}_{A_+^*}(\varepsilon) = \operatorname{sepd}(A_+^*)$, and taking the supremum over annihilator nets in B_P gives $\operatorname{asepd}(P) \le \langle\, \operatorname{sepd}(A_+^*) \,/\, C \operatorname{sepd}(A_+^*) \,\rangle$.

For the lower bound, P is not essential, so proposition 2.6 and theorem 2.11(i) give $\operatorname{asepd}(P) \ge \operatorname{sepd}(P^*) \ge \operatorname{sepd}(A_+^*)$. In the metric bracket the two bounds coincide. □

Comparing this with theorem 2.17 shows that the two layers must be kept apart. Over $c_0(\Lambda)$ a non-essential projective module has $\operatorname{sepd}(P) = 0$ and $\operatorname{asepd}(P) = \frac{1}{2}$: every annihilator vector lies inside P_{ess}, and the whole defect is carried by annihilator nets in B_P that stay at positive distance from P_{ann}.

2.4 The annihilator part of an injective module

Interleaved from [B], in this order, before the material below: proposition `MetTopRelInjModCompIdealAnnihCompl`, corollary `MetTopRelInjModAnnihCompl`, proposition `MetTopInjModAnnihPartIsInjBanSp`.

Injectivity and flatness are governed by the dual constant instead, and by the same mechanism: an annihilator direction of a dual module is transported into the module under test. For injectivity the transport is an extension, for flatness a tensor factor. Lemma 2.2 is the pointwise content of the averaging projection built in the proof of proposition `MetTopRelInjModCompIdealAnnihCompl`: the annihilator part is not merely complemented, each of its vectors stands at a definite distance from the essential part.

Proposition 2.21. *Let J be a $\langle$ metrically / C-topologically $\rangle$ injective right A-module, let X be a left A-module and let $f \in (X_{ess})^\perp$ with $\|f\| = 1$. Then*

$$\operatorname{sep}_J(f_0) \le \langle\, 1 \,/\, C \,\rangle \operatorname{sep}_{X^*}(f)\, \|f_0\| \qquad (f_0 \in J_{ann});$$

in particular $\operatorname{sep}_J(f_0) \le \langle\, \operatorname{sepd}(A_+^*) \,/\, C \operatorname{sepd}(A_+^*) \,\rangle \|f_0\|$.

Proof. Since $f \in (X_{ess})^\perp = (X^*)_{ann}$, the line $\mathbb{C}f \subset X^*$ is a closed submodule on which the action is zero, and the inclusion $\xi : \mathbb{C}f \to X^*$ is an isometric A-morphism, hence 1-topologically injective. The map $\varphi : \mathbb{C}f \to J$, $\lambda f \mapsto \lambda f_0$ is well defined and linear with $\|\varphi\| = \|f_0\|$, and it is

an A-morphism: $\varphi\big((\lambda f)\cdot a\big) = \varphi(0) = 0$ while $\varphi(\lambda f)\cdot a = \lambda(f_0\cdot a) = 0$ because $f_0 \in J_{ann}$. By definition `MetCTopInjMod`, with $c = 1$, injectivity of J extends φ along ξ: there is an A-morphism $\psi : X^* \to J$ with $\psi(f) = f_0$ and $\|\psi\| \le \langle\, \|f_0\| \;/\; C\|f_0\| \,\rangle$.

A morphism carries essential parts into essential parts, since $\psi(g\cdot a) = \psi(g)\cdot a \in \operatorname{span}(JA)$ and J_{ess} is closed; so $\psi\big((X^*)_{ess}\big) \subset J_{ess}$ and, for every $g \in (X^*)_{ess}$,

$$\operatorname{sep}_J(f_0) \le \|f_0 - \psi(g)\| = \|\psi(f) - \psi(g)\| \le \|\psi\|\|f - g\|.$$

Taking the infimum over g turns the right-hand side into $\|\psi\|\operatorname{sep}_{X^*}(f)$, which is the displayed bound. For the last assertion take $(X, f) = (A_+, \varepsilon)$, admissible by lemma 2.10, which also gives $\operatorname{sep}_{A_+^*}(\varepsilon) = \operatorname{sepd}(A_+^*)$. □

Corollary 2.22. *Let A have a right approximate identity. Then in every metrically injective right A-module J,*

$$\operatorname{sep}_J(f_0) = \operatorname{sepd}(A_+^*)\|f_0\| = \frac{\|f_0\|}{1 + c_{\mathrm{rai}}(A)} \qquad (f_0 \in J_{ann}),$$

and $\operatorname{sepd}(A_+^*)\|f_0\| \le \operatorname{sep}_J(f_0) \le C\operatorname{sepd}(A_+^*)\|f_0\|$ *in every C-topologically injective one. The value is attained on A_+^* at $f_0 = \varepsilon$. If A has no right approximate identity, then $J_{ann} \subset J_{ess}$ in every topologically injective right A-module.*

Proof. The upper bounds are proposition 2.21 with $(X, f) = (A_+, \varepsilon)$. For the lower bound, let $\epsilon > 0$ and let (e_ν) be a right approximate identity bounded by $K = c_{\mathrm{rai}}(A) + \epsilon$; lemma 2.2 in its right-hand form, applied to J, gives $\operatorname{sep}_J(f_0) \ge \|f_0\|/(1 + K)$, and letting $\epsilon \to 0$ and using theorem 2.11(*ii*) gives $\operatorname{sep}_J(f_0) \ge \operatorname{sepd}(A_+^*)\|f_0\|$. In the metric bracket the two bounds agree.

The value is attained: A_+ is free of rank one, hence metrically projective by proposition `MetCTopFreeMod`, so A_+^* is metrically injective by proposition `DualMetTopProjIsMetrInj`, and its annihilator part is the line $\mathbb{C}\varepsilon$ with $\|\varepsilon\| = 1$, on which lemma 2.10 gives $\operatorname{sep}_{A_+^*}(\varepsilon) = \operatorname{sepd}(A_+^*)$.

Finally, suppose A has no right approximate identity; then $A \ne \{0\}$, since the zero net is one for the zero algebra, so corollary 2.13 gives $\operatorname{sepd}(A_+^*) = 0$ and the topological upper bound reads $\operatorname{sep}_J(f_0) \le 0$ for every $f_0 \in J_{ann}$. As J_{ess} is closed, this says $J_{ann} \subset J_{ess}$. □

Corollary 2.23. *Let $J \ne \{0\}$ be an annihilator right A-module which is $\langle$ metrically / C-topologically $\rangle$ injective. Then $\langle$ $A = \{0\}$ / A has a right $(C-1)$-bounded approximate identity $\rangle$.*

Proof. Since J is a non-zero annihilator module, $J_{ess} = \{0\}$ and $J_{ann} = J$, so $\operatorname{sep}_J(f_0) = \|f_0\|$ for every $f_0 \in J$. Applying proposition 2.21 at some $f_0 \ne 0$ with $(X, f) = (A_+, \varepsilon)$ gives $\operatorname{sepd}(A_+^*) \ge \langle\, 1 \;/\; 1/C \,\rangle$. In the metric bracket this says $\operatorname{sepd}(A_+^*) \ge 1$, contradicting the ceiling $\frac{1}{2}$ of lemma 2.12 unless the hypothesis of that lemma fails, that is unless $A = \{0\}$. In the topological bracket it says $1/(1 + c_{\mathrm{rai}}(A)) \ge 1/C$, that is $c_{\mathrm{rai}}(A) \le C - 1$; by theorem 2.11(*iii*) this bound is attained, so A has a right approximate identity bounded by $C - 1$. □

Corollary 2.22 is pointwise, and taking the supremum over $B_{J_{ann}}$ turns it into a statement about the defect itself: on metrically injective modules sepd is two-valued.

Corollary 2.24. *Let J be a $\langle$ metrically / C-topologically $\rangle$ injective right A-module. If J is faithful, then* $\operatorname{sepd}(J) = 0$*; otherwise*

$$\langle\, \operatorname{sepd}(J) = \operatorname{sepd}(A_+^*) \;/\; \operatorname{sepd}(A_+^*) \le \operatorname{sepd}(J) \le C\operatorname{sepd}(A_+^*) \,\rangle,$$

so that on metrically injective modules the separation defect takes only the two values 0 *and* $\operatorname{sepd}(A_+^*)$*. Consequently, in the metric case:*

(i) $\operatorname{sepd}(P^*) = \operatorname{sepd}(A_+^*)$ *for every left A-module P that is metrically projective and not essential — the cofree modules of proposition 2.15 being the case $P = A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$;*

(ii) $\operatorname{sepd}\big(\bigoplus_\infty \{J_\gamma : \gamma \in \Gamma\}\big) = \sup_\gamma \operatorname{sepd}(J_\gamma)$ *for every family of metrically injective right A-modules, where proposition 2.9(ii) gives only an inequality once the family is infinite.*

Proof. Faithfulness means $J_{ann} = \{0\}$, and then $\operatorname{sepd}(J) = \operatorname{sep}_J(0) = 0$. Assume $J_{ann} \neq \{0\}$, so that J_{ann} contains a vector of norm one. If A has a right approximate identity, corollary 2.22 gives the pointwise equality, respectively the two pointwise bounds, and taking the supremum over $f_0 \in B_{J_{ann}}$ gives the displayed bracket, the supremum being attained at a vector of norm one. If A has no right approximate identity then $A \neq \{0\}$, so corollary 2.13 gives $\operatorname{sepd}(A_+^*) = 0$, while the last clause of corollary 2.22 gives $J_{ann} \subset J_{ess}$ and hence $\operatorname{sepd}(J) = 0$; both brackets then read $0 = 0$.

(*i*) Let P be metrically projective and not essential. Its dual is metrically injective by proposition `DualMetTopProjIsMetrInj`, and $(P^*)_{ann} = (P_{ess})^\perp$ is non-zero precisely because $P_{ess} \neq P$; so the main clause gives $\operatorname{sepd}(P^*) = \operatorname{sepd}(A_+^*)$. The cofree modules are the case $P = A_+ \mathbin{\widehat{\otimes}} \ell_1(\Lambda)$, which is metrically projective by proposition `MetCTopFreeMod` and whose dual is $\mathcal{B}(A_+, \ell_\infty(\Lambda))$ by the identification made in the proof of proposition 2.15.

(*ii*) The sum is metrically injective by proposition `MetTopInjModProd`, and each J_γ is a 1-retract of it, so lemma 2.8 gives $\geq$. For the reverse, apply the main clause to the sum itself: by proposition 2.9 its annihilator part is $\bigoplus_\infty \{(J_\gamma)_{ann} : \gamma \in \Gamma\}$, which is non-zero exactly when some $(J_\gamma)_{ann}$ is. If none is, both sides are 0. If some is, the left-hand side equals $\operatorname{sepd}(A_+^*)$, and so does $\operatorname{sepd}(J_\gamma)$ for that γ. □

2.5 The annihilator part of a flat module

Interleaved from [B], in this order, before the material below: proposition `MetTopRelFlatModCompIdealPartCompl`, corollary `MetTopRelFlatModPartCompl`, proposition `MetTopFlatModEssQuotIsFlatBanSp`.

The engine here is not the separation defect but a statement about the module tensor product; the defect appears in its corollaries.

Proposition 2.25. *Let F be a left A-module.*

(i) For every right A-module Y, every $y_0 \in Y_{ann}$ and every $x \in F$,

$$\|y_0 \otimes x\|_{Y \widehat{\otimes}_A F} \leq \|y_0\| \operatorname{sep}_F(x).$$

(ii) If F is $\langle$ metrically / C-topologically $\rangle$ flat, then

$$\big\langle \|y_0 \otimes x\|_{Y \widehat{\otimes}_A F} = \|y_0\| \operatorname{sep}_F(x) \;/\; \|y_0\| \operatorname{sep}_F(x) \leq C \|y_0 \otimes x\|_{Y \widehat{\otimes}_A F} \big\rangle.$$

(iii) Consequently, if F is $\langle$ metrically / C-topologically $\rangle$ flat, then for every left A-module X, every $f \in (X_{ess})^\perp$ with $\|f\| = 1$, every $x \in F$ and all finite families $g_1, \ldots, g_n \in X^*$ and $a_1, \ldots, a_n \in A$,

$$\operatorname{sep}_F(x) \leq \langle\, 1 \;/\; C \,\rangle \cdot \Big(\big\| f - \textstyle\sum_i g_i \cdot a_i \big\| \|x\| + \sum_i \|g_i\| \|a_i \cdot x\| \Big).$$

Proof. (i) By the law of adjoint associativity there is an isometric isomorphism $\mathcal{B}(Y \mathbin{\widehat{\otimes}_A} F, \mathbb{C}) \underset{\mathbf{Ban}_1}{\cong} \mathcal{B}_A(Y, F^*)$, under which a functional Φ corresponds to the morphism T with $\langle \Phi, y \otimes x \rangle = \langle Ty, x \rangle$. The norm of a vector being the supremum of the moduli of the unit functionals at it,

$$\|y_0 \otimes x\|_{Y \mathbin{\widehat{\otimes}_A} F} = \sup\{|\langle Ty_0, x\rangle| : T \in \mathcal{B}_A(Y, F^*),\ \|T\| \le 1\}.$$

Fix such a T. It is a morphism into the right module F^*, so $(Ty_0)\cdot a = T(y_0 \cdot a) = 0$ for every $a \in A$; that is, $Ty_0 \in (F^*)_{ann} = (F_{ess})^\perp$ with $\|Ty_0\| \le \|y_0\|$. Hence $|\langle Ty_0, x\rangle| = |\langle Ty_0, x - z\rangle| \le \|y_0\|\|x - z\|$ for every $z \in F_{ess}$, and the infimum over z is $\|y_0\| \operatorname{sep}_F(x)$.

(ii) Both sides are positively homogeneous of degree one in y_0 and vanish at $y_0 = 0$, so we may assume $\|y_0\| = 1$. The line $Y_0 := \mathbb{C}y_0$ is a closed submodule of Y on which the action is zero, and the inclusion $\xi : Y_0 \to Y$ is an isometric A-morphism, hence also 1-topologically injective, so it belongs to the test class of definition `MetCTopFlatMod` in both brackets with $c = 1$; flatness of F therefore makes $\xi \mathbin{\widehat{\otimes}_A} 1_F$ $\langle$ isometric / C-topologically injective $\rangle$. Now identify the source. The assignment $y_0 \otimes x \mapsto x$ extends to an isometric isomorphism $Y_0 \mathbin{\widehat{\otimes}} F \to F$, since Y_0 is one-dimensional and spanned by a unit vector; under it the subspace by which one divides to form $Y_0 \mathbin{\widehat{\otimes}_A} F$, namely $\operatorname{cl}(\operatorname{span}\{y_0 \cdot a \otimes x - y_0 \otimes a \cdot x\}) = \operatorname{cl}(\operatorname{span}\{y_0 \otimes a \cdot x\})$, corresponds to F_{ess}. Passing to quotients gives an isometric isomorphism $Y_0 \mathbin{\widehat{\otimes}_A} F \underset{\mathbf{Ban}_1}{\cong} F/F_{ess}$ carrying $y_0 \otimes x$ to the coset $x + F_{ess}$, so that $\|y_0 \otimes x\|_{Y_0 \mathbin{\widehat{\otimes}_A} F} = \operatorname{sep}_F(x)$. Feeding this through $\xi \mathbin{\widehat{\otimes}_A} 1_F$ gives (ii) in both brackets.

(iii) Apply (ii) with $(Y, y_0) = (X^*, f)$, legitimate because $f \in (X_{ess})^\perp = (X^*)_{ann}$ and $\|f\| = 1$; it gives $\operatorname{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \|f \otimes x\|_{X^* \mathbin{\widehat{\otimes}_A} F}$. In $X^* \mathbin{\widehat{\otimes}_A} F$ the balancing relation $(g \cdot a) \otimes x = g \otimes (a \cdot x)$ holds, so $f \otimes x = \big(f - \sum_i g_i \cdot a_i\big) \otimes x + \sum_i g_i \otimes (a_i \cdot x)$, whence the displayed bound by the triangle inequality and $\|u \otimes v\| \le \|u\|\|v\|$. ☐

Corollary 2.26. *Let F be a $\langle$ metrically / C-topologically $\rangle$ flat left A-module. Then for every $x \in F$, every $a \in A \setminus \{0\}$ and every $0 < \gamma \le 1$,*

$$\operatorname{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \Big(\max\{\gamma, 1 - \gamma\}\|x\| + \gamma \frac{\|a \cdot x\|}{\|a\|} \Big);$$

with $\gamma = \frac{1}{2}$, $2 \operatorname{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \big(\|x\| + \|a \cdot x\|/\|a\|\big)$.

Proof. The statement quantifies over $a \in A \setminus \{0\}$, so it is vacuous for $A = \{0\}$ and we may assume $A \neq \{0\}$; lemma 2.12 is then available. Apply proposition 2.25(iii) with $(X, f) = (A_+, \varepsilon)$ — admissible by lemma 2.10 — and with the single term $g_1 := (\gamma/\|a\|)\varphi$, $a_1 := a$, where $\varphi \in A_+^*$ satisfies $\|\varphi\| = 1$ and $\varphi(a/\|a\|) = 1$. Then $g_1 \cdot a_1 = \gamma\, \varphi \cdot (a/\|a\|)$, so lemma 2.12 applied to the norm-one element $a/\|a\|$ bounds $\|\varepsilon - g_1 \cdot a_1\|$ by $\max\{\gamma, 1-\gamma\}$, while $\|g_1\|\|a_1 \cdot x\| = \gamma\|a \cdot x\|/\|a\|$. Substituting the two bounds gives the display, and $\max\{\gamma, 1-\gamma\} = \frac{1}{2}$ at $\gamma = \frac{1}{2}$. ☐

Taking the infimum over $a \neq 0$ in the second display gives the form in which the estimate holds at every vector of F, not only the annihilator ones:

$$2 \operatorname{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \big(\|x\| + \operatorname{sng}(x)\big) \qquad (x \in F),$$

and $\gamma = \frac{1}{2}$ is the optimal choice there because $\operatorname{sng}(x) \le \|x\|$ always. This is not subsumed by the annihilator estimate that follows, nor does it subsume it: the constant $\frac{1}{2}$ is the value of $\operatorname{sepd}(A_+^*)$ only when A has a contractive right approximate identity.

Corollary 2.27. *Let F be $\langle$ metrically / C-topologically $\rangle$ flat and $x \in F_{ann}$. Then $\mathrm{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \mathrm{sep}_{X^*}(f)\|x\|$ for every admissible pair (X, f), that is, for every left A-module X and every $f \in (X_{ess})^\perp$ with $\|f\| = 1$; and the best choice is (A_+, ε):*

$$\mathrm{sep}_F(x) \le \langle\, \mathrm{sepd}(A_+^*) \,/\, C\,\mathrm{sepd}(A_+^*) \,\rangle \|x\|.$$

Proof. For $x \in F_{ann}$ every term $\|a_i \cdot x\|$ in proposition 2.25(iii) vanishes, so that proposition reads $\mathrm{sep}_F(x) \le \langle\, 1 \,/\, C \,\rangle \|f - \sum_i g_i \cdot a_i\| \|x\|$ for all finite families. The elements $\sum_i g_i \cdot a_i$ are exactly the members of $\mathrm{span}\{g \cdot a\}$, which is dense in $(X^*)_{ess}$; taking the infimum over them replaces the norm by $\mathrm{dist}\big(f, (X^*)_{ess}\big) = \mathrm{sep}_{X^*}(f)$.

For the best choice, theorem 2.11(i) gives $\mathrm{sep}_{X^*}(f) \ge \|f\|/(1 + c_{\mathrm{rai}}(A)) = \mathrm{sepd}(A_+^*)$ for every admissible pair, so none yields a coefficient smaller than $\mathrm{sepd}(A_+^*)$; and the pair (A_+, ε) attains it, since $\mathrm{sep}_{A_+^*}(\varepsilon) = \mathrm{sepd}(A_+^*)$ by lemma 2.10. No hypothesis on A is used. □

Corollary 2.28. *Let $F \neq \{0\}$ be an annihilator $\langle$ metrically / C-topologically $\rangle$ flat A-module. Then $\langle\, A = \{0\} \,/\, A$ has a right $(C-1)$-bounded approximate identity $\rangle$.*

Proof. Since F is a non-zero annihilator module, $F_{ess} = \{0\}$ and $F_{ann} = F$, so $\mathrm{sep}_F(x) = \|x\|$ for every $x \in F$; applying corollary 2.27 at some $x \neq 0$ gives $\mathrm{sepd}(A_+^*) \ge \langle\, 1 \,/\, 1/C \,\rangle$. In the metric bracket this says $\mathrm{sepd}(A_+^*) \ge 1$, contradicting the ceiling $\frac{1}{2}$ of lemma 2.12 unless the hypothesis of that lemma fails, that is unless $A = \{0\}$. In the topological bracket theorem 2.11(ii) turns it into $c_{\mathrm{rai}}(A) \le C - 1$, and (iii) of the same theorem says the bound is attained. □

Corollary 2.29. *Let F be a $\langle$ metrically / C-topologically $\rangle$ flat left A-module. Then, with no hypothesis on A,*

$$\frac{\|x\|}{1 + c_{\mathrm{lai}}(A)} \;\le\; \mathrm{sep}_F(x) \;\le\; \langle\, \mathrm{sepd}(A_+^*) \,/\, C\,\mathrm{sepd}(A_+^*) \,\rangle \|x\| \qquad (x \in F_{ann}).$$

If moreover $c_{\mathrm{lai}}(A) \le c_{\mathrm{rai}}(A)$, the metric bracket collapses to the equality

$$\mathrm{sep}_F(x) = \mathrm{sepd}(A_+^*)\|x\| = \frac{\|x\|}{1 + c_{\mathrm{rai}}(A)} \qquad (x \in F_{ann}),$$

which holds in particular when A is commutative, when A is unital, where the value is $\|x\|/(1 + \|e_A\|)$, and when $A \neq \{0\}$ has a contractive two-sided approximate identity, where it is $\frac{1}{2}\|x\|$. Conversely, if some metrically flat left A-module has a non-zero annihilator vector, then $c_{\mathrm{rai}}(A) \le c_{\mathrm{lai}}(A)$.

Proof. The upper bound is corollary 2.27. For the lower bound, if A has a left approximate identity then for every $\epsilon > 0$ there is one bounded by $K = c_{\mathrm{lai}}(A) + \epsilon$, and lemma 2.2 applied to the left module F gives $\mathrm{sep}_F(x) \ge \|x\|/(1 + K)$; let $\epsilon \to 0$. If A has none then $c_{\mathrm{lai}}(A) = +\infty$ and the bound is vacuous.

Suppose $c_{\mathrm{lai}}(A) \le c_{\mathrm{rai}}(A)$. Then $\|x\|/(1 + c_{\mathrm{lai}}(A)) \ge \|x\|/(1 + c_{\mathrm{rai}}(A)) = \mathrm{sepd}(A_+^*)\|x\|$ by theorem 2.11(ii), so in the metric bracket the lower bound is at least the upper one and the two meet.

The three instances are the elementary relations recorded in subsection `SubSectionBanachAlgebrasAndTheirMo` a commutative algebra has $c_{\mathrm{lai}}(A) = c_{\mathrm{rai}}(A)$; a unital one has $c_{\mathrm{lai}}(A) = c_{\mathrm{rai}}(A) = \|e_A\|$, the zero algebra included, with $e_A = 0$ and value $\|x\|$; and an algebra $A \neq \{0\}$ with a contractive two-sided approximate identity has $c_{\mathrm{lai}}(A) = c_{\mathrm{rai}}(A) = 1$, since such a net is bounded by 1 on both sides

while every approximate identity of a non-zero algebra is bounded below by 1. The hypothesis $A \neq \{0\}$ is not removable in the third instance: the zero algebra has the zero net as a contractive two-sided approximate identity, yet $c_{\mathrm{lai}} = c_{\mathrm{rai}} = 0$ there and the value is $\|x\|$.

For the converse, let F be metrically flat with some $x \in F_{ann}$, $x \neq 0$. Reading the displayed sandwich at that single x and dividing by $\|x\|$ gives $1/(1 + c_{\mathrm{lai}}(A)) \leq \mathrm{sepd}(A_+^*) = 1/(1 + c_{\mathrm{rai}}(A))$, that is $c_{\mathrm{rai}}(A) \leq c_{\mathrm{lai}}(A)$. □

On the asymptotic layer the estimate simplifies: the master inequality bounds sep_F at every vector, not only the annihilator ones, so it survives the passage to nets.

Proposition 2.30. *Let $A \neq \{0\}$ and let X be a $\langle$ metrically / C-topologically $\rangle$ flat left A-module. Then* $\mathrm{asepd}(X) \leq \langle\, 1/2 \;/\; C/2 \,\rangle$*; equivalently, every topological flatness constant of X satisfies $C \geq 2\,\mathrm{asepd}(X)$.*

Proof. Since $A \neq \{0\}$ we may fix $a_0 \in A$ with $\|a_0\| = 1$. Let $(x_\nu)_{\nu \in N}$ be an annihilator net in B_X; corollary 2.26 at $\gamma = \frac{1}{2}$ and $a = a_0$ gives $2\,\mathrm{sep}_X(x_\nu) \leq \langle\, 1 \;/\; C \,\rangle\big(1 + \|a_0 \cdot x_\nu\|\big)$. Take the limit superior, use $\|a_0 \cdot x_\nu\| \to 0$, and take the supremum over annihilator nets in B_X. □

The metric bracket is sharp on free modules, which are metrically flat by propositions `MetCTopFreeMod` and `MetTopProjIsMetTopFlat`: by proposition 2.14 they have $\mathrm{asepd} = \mathrm{sepd}(A_+^*)$, and that equals $\frac{1}{2}$ exactly when A has a contractive right approximate identity.

Corollary 2.31. *Let $A \neq \{0\}$ have a right approximate identity. Then every non-essential $\langle$ metrically / C-topologically $\rangle$ flat left A-module X satisfies*

$$\frac{1}{1 + c_{\mathrm{rai}}(A)} \leq \mathrm{asepd}(X) \leq \langle\, 1/2 \;/\; C/2 \,\rangle.$$

In particular, over an algebra with a contractive right approximate identity the asymptotic separation defect of a metrically flat module takes exactly two values: $\mathrm{asepd}(X) = 0$ *if X is essential and* $\mathrm{asepd}(X) = \frac{1}{2}$ *otherwise.*

Proof. The upper bound is proposition 2.30. For the lower one, X is not essential, so proposition 2.6 and theorem 2.11(i), (ii) give $\mathrm{asepd}(X) \geq \mathrm{sepd}(X^*) \geq \mathrm{sepd}(A_+^*) = 1/(1 + c_{\mathrm{rai}}(A))$. If A has a contractive right approximate identity then $c_{\mathrm{rai}}(A) = 1$ by the first clause of corollary 2.13, so for non-essential metrically flat X the display reads $\frac{1}{2} \leq \mathrm{asepd}(X) \leq \frac{1}{2}$; and if X is essential then sep_X vanishes identically. □

Corollary 2.32. *Let $A \neq \{0\}$. A left A-module of asymptotic separation defect* 1 *is never metrically flat, and its topological flatness constants satisfy $C \geq 2$.*

Proof. If X were metrically flat, proposition 2.30 would give $\mathrm{asepd}(X) \leq \frac{1}{2}$, contradicting $\mathrm{asepd}(X) = 1$; and if X is C-topologically flat the same proposition gives $1 \leq C/2$.

The hypothesis $A \neq \{0\}$ is needed, and $\mathbb{C}$ is the witness. Over the zero algebra every module is an annihilator module, so $\mathbb{C}_{ess} = \{0\}$, $\mathrm{sep}_{\mathbb{C}} = \|\cdot\|$ and $\mathrm{asepd}(\mathbb{C}) = 1$; on the other hand, as recorded in subsection `SubSectionMetricAndTopologicalFlatness`, modules over the zero algebra are Banach spaces and metric flatness for them is Grothendieck's L_1 condition [`GrothMetrProjFlatBanSp`, theorem 1], which $\mathbb{C}$ satisfies as the L_1-space of a one-point measure space. So $\mathbb{C}$ is a metrically flat module of asymptotic separation defect 1 over $A = \{0\}$. □

2.6 Homologically trivial annihilator modules

This subsection is the present subsection 2.2.1 of [B], moved here unchanged in content and order: proposition `AnnihCModIsRetAnnihMod`, proposition `MetTopProjModCCharac`, proposition `MetTopProjOfAnnihModCh` proposition `MetTopInjModCCharac`, proposition `MetTopInjOfAnnihModCharac`, proposition `MetTopFlatAnnihModCh` and the proposition asserting that no Banach algebra makes all of its modules flat, which is unlabelled in [B] and needs a label before it can be cited (`NoAlgWithAllModFlat` is proposed in Part 4). *What changes is the proof of the algebra clause of each* $(i) \Rightarrow (ii)$ *half, and two corollaries are added.* The three characterisations then have one proof with two constants: $\mathrm{sepd}(A_+)$ for the projective case and $\mathrm{sepd}(A_+^*)$ for the other two.

The algebra clause of proposition `MetTopProjModCCharac`$(i) \Rightarrow (ii)$ **and of proposition** `MetTopProjOfAnnihl` (ii)**.** Let P be a non-zero annihilator A-module which is $\langle$ metrically / C-topologically $\rangle$ projective; the case $P = \mathbb{C}$ is proposition `MetTopProjModCCharac`. By corollary 2.18, $\langle$ $A = \{0\}$ / $c_{\mathrm{ri}}(A) \leq C-1$ $\rangle$. This is one application of theorem 2.17 to the single lifting problem $\varepsilon : A_+ \to \mathbb{C}$, in place of the idempotent argument of proposition `MetTopProjCycModCharac`, and by corollary 2.19 it says more than the existence of the identity: over an algebra without a right identity every annihilator vector of a projective module already lies in its essential part.

One caveat on the topological bracket. The clause as printed in [B] asks for a right identity of norm at most $C-1$, and the separation estimate gives $c_{\mathrm{ri}}(A) \leq C-1$, that is, a right identity of norm at most $C-1+\epsilon$ for every $\epsilon > 0$. The two are not the same, because the infimum defining $c_{\mathrm{ri}}(A)$ need not be attained: the right identities of A form a closed coset $u + \{z \in A : Az = \{0\}\}$, and closed subspaces of a Banach space are not in general proximinal. This is exactly where the projective constant differs from the dual one, whose infimum *is* attained (theorem 2.11(iii)). Either restate the clause as $c_{\mathrm{ri}}(A) \leq C-1$ — which is the invariant form, and which we recommend — or keep the attained form and with it the appeal to proposition `MetTopProjCycModCharac`, which is legal here since that proposition lies in section 2.1.

The algebra clause of proposition `MetTopInjModCCharac` **and of proposition** `MetTopInjOfAnnihModCharac`$(i$ (ii)**.** Let $J \neq \{0\}$ be an annihilator right A-module which is $\langle$ metrically / C-topologically $\rangle$ injective. By corollary 2.23, $\langle$ $A = \{0\}$ / A has a right $(C-1)$-bounded approximate identity $\rangle$; here the bound *is* attained, by theorem 2.11(iii). This is three lines and replaces the passage through metric and topological flatness of A_+/A.

The algebra clause of proposition `MetTopFlatAnnihModCharac`$(i) \Rightarrow (ii)$**.** Identical, with corollary 2.28 in place of corollary 2.23.

The constant $C-1$ is moreover optimal, and the least admissible C is the reciprocal of the dual constant.

Corollary 2.33. *Let $J \neq \{0\}$ be an annihilator right A-module which is metrically injective as a Banach space. Then*

$$\inf\{C : J \text{ is a } C\text{-topologically injective } A\text{-module}\} = \frac{1}{\mathrm{sepd}(A_+^*)} = 1 + c_{\mathrm{rai}}(A),$$

and the infimum is attained; it is infinite exactly when A has no right approximate identity. The same holds, with the same proof, for an annihilator A-module $F \neq \{0\}$ that is metrically flat as a Banach space and for the constant of proposition `MetTopFlatAnnihModCharac`*.*

Proof. By corollary 2.23 every admissible C satisfies $C \geq 1 + c_{\mathrm{rai}}(A) = 1/\operatorname{sepd}(A_+^*)$, which is also the assertion about finiteness. For the upper bound let $c_{\mathrm{rai}}(A) < +\infty$ and put $C := 1 + c_{\mathrm{rai}}(A)$. By theorem 2.11(*iii*) the algebra A has a right $(C-1)$-bounded approximate identity; and J, being metrically injective as a Banach space, is 1-topologically injective as a Banach space by proposition `MetInjIsOneTopInj` read over the zero algebra, hence C-topologically injective as a Banach space for every $C \geq 1$. These are exactly the hypotheses of proposition `MetTopInjOfAnnihModCharac` in the direction $(ii) \Rightarrow (i)$, whose conclusion is that J is C-topologically injective as an A-module. The flat statement is the same argument with proposition `MetTopFlatAnnihModCharac` in place of proposition `MetTopInjOfAnnihModCharac`, a metrically flat Banach space being an $\mathscr{L}^g_{1,1}$-space and hence an $\mathscr{L}^g_{1,C}$-space for every $C \geq 1$. □

Finally, the asymptotic layer carries the same three characterisations one step further: propositions 2.20 and 2.30 with corollaries 2.31 and 2.32 say that over an algebra with a contractive right approximate identity a metrically flat module has asepd $\in \{0, \frac{1}{2}\}$, and that a module of asymptotic separation defect 1 over a non-zero algebra is never metrically flat. That last statement is what the measure algebra needs, and it is proved in Part 3.

3 For chapter 3

3.1 For subsection `SectionPreliminariesOnHarmonicAnalysis`, replacing the sentences on the Lebesgue decomposition of $M(G)$

Delete the assertions that $M(G)$ is a coproduct of $M_a(G)$ and $M_s(G)$ in $L_1(G)$-$\mathbf{mod}_1$, that $M_s(G)$ is a subalgebra of $M(G)$, and that $M_s(G)$ is an annihilator $L_1(G)$-module. All three fail for non-discrete G, because $M_s(G)$ is then not even an $L_1(G)$-submodule of $M(G)$: points are m_G-null, so $\delta_x \in M_s(G)$ for every $x \in G$, while $gm_G * \delta_x$ is non-zero and absolutely continuous for every $g \neq 0$ in $L_1(G)$; hence $L_1(G) * M_s(G) \not\subset M_s(G)$ and $M_s(G) * L_1(G) \neq \{0\}$. Nor is $M_s(G)$ a subalgebra: Wiener and Wintner constructed on $\mathbb{T}$ a singular probability measure μ with $\widehat{\mu}(n) = O(|n|^{-1/2+\epsilon})$, and then μ^{*4} has absolutely convergent Fourier series and is a non-zero absolutely continuous measure.

Insert instead: $M_a(G)$ is a closed two-sided ideal of $M(G)$ with $M_a(G) \underset{M(G)\text{-}\mathbf{mod}_1}{\cong} L_1(G)$; the Lebesgue decomposition $M(G) \underset{\mathbf{Ban}_1}{\cong} M_a(G) \bigoplus_1 M_s(G)$ is an isomorphism of Banach spaces and of Banach spaces only; and $M(G)_{ess} = M_a(G)$, so that $M(G)/M(G)_{ess} \underset{\mathbf{Ban}_1}{\cong} M_s(G)$, this last being proved in proposition 3.1 below. That is what the proofs of proposition `L1MetTopProjAndMetrFlatOfMeasAlg`(*iii*) and of proposition `MeasAlgIsL1TopFlat` already use, and all that they need; the opening paragraph of the latter, which proves $M(G)_{ess} = M_a(G)$ a second time, can be cut and replaced by a citation.

One sentence is worth adding at the point of deletion, because it is how the error arose. Subsection `SubSectionCOSModules` correctly asserts that $M_s(S, \mu)$ *is* a closed $C_0(S)$-submodule of $M(S)$: there the action is pointwise multiplication, which preserves singularity with respect to μ. Here the action is convolution, which does not.

3.2 For subsection `SubSectionL1GModules`

The following proposition goes at the head of the subsection. It is the supporting statement for the correction above and for the replacement proof that follows it.

Proposition 3.1. *Let G be a non-discrete locally compact group and regard $M(G)$ as a left $L_1(G)$-module. Then $M(G)_{ess} = M_a(G)$ and $M(G)_{ann} = \{0\}$, and*

$$\operatorname{asepd}(M(G)) = 1, \qquad \operatorname{sepd}(M(G)^*) = 1, \qquad \operatorname{sepd}(M(G)) = 0.$$

Proof. Points are Haar-null. By left invariance $m_G(\{x\}) = m_G(\{e_G\}) =: c$ for every $x \in G$. If $c > 0$, take a compact neighbourhood W of e_G, so that $m_G(W) < +\infty$; any n distinct points of W give $m_G(W) \geq nc$, so W is finite, and a finite neighbourhood of e_G in a Hausdorff group contains a finite open set, which makes $\{e_G\}$ open and G discrete. Hence $c = 0$ and $\delta_x \in M_s(G)$ for every $x \in G$.

The essential part. On the one hand $L_1(G) * M(G) \subset M_a(G)$, a convolution with an absolutely continuous measure being absolutely continuous; on the other hand $M_a(G)$ is isometrically isomorphic to $L_1(G)$ as a module and $L_1(G)$ has a contractive approximate identity, so $M_a(G)$ is essential. Hence $M(G)_{ess} = M_a(G)$. The annihilator part is zero: if $f * \mu = 0$ for every $f \in L_1(G)$, take $f = \chi_U / m_G(U)$ over a shrinking neighbourhood base at e_G, so that $f * \mu \to \mu$ weak* and $\mu = 0$. Consequently $B_{M(G)_{ann}} = \{0\}$ and $\operatorname{sepd}(M(G)) = 0$: the measure algebra is a witness for the strictness of the first inequality of proposition 2.6, and it realises the maximal asymptotic defect without a single annihilator vector.

The separation of a difference of point masses. For $z \neq e_G$ put $\mu_z := \frac{1}{2}(\delta_{e_G} - \delta_z)$. The two point masses are mutually singular, so $\|\mu_z\| = 1$, and $\mu_z \in M_s(G)$; the $\bigoplus_1$-decomposition gives $\|\mu_z - \nu\| = \|\mu_z\| + \|\nu\| \geq 1$ for every $\nu \in M_a(G)$, whence $\operatorname{sep}_{M(G)}(\mu_z) = 1$.

An annihilator net at separation one. Let N be a neighbourhood base at e_G, directed by reverse inclusion. Since G is not discrete, no $V \in N$ reduces to $\{e_G\}$, so we may choose $z_V \in V \setminus \{e_G\}$. For $g \in L_1(G)$ one has $g * \mu_{z_V} = \frac{1}{2}(g - g * \delta_{z_V})$, and $\|g - g * \delta_z\|_1 \to 0$ as $z \to e_G$ by the norm continuity of translation in $L_1(G)$ together with the continuity of the modular function; so $(\mu_{z_V})_{V \in N}$ is an annihilator net in $B_{M(G)}$ with $\operatorname{sep}_{M(G)}(\mu_{z_V}) = 1$ at every index, and $\operatorname{asepd}(M(G)) \geq 1$. As $\operatorname{asepd} \leq 1$ always, this is an equality.

The dual. Every $\mu \in M(G)$ has at most countably many atoms and $\sum_x |\mu(\{x\})| \leq \|\mu\|$, so $f(\mu) := \mu(\{e_G\}) - \sum_{x \neq e_G} \mu(\{x\})$ is a well-defined linear functional with $\|f\| \leq 1$. It vanishes on $M_a(G)$, an absolutely continuous measure having no atoms, that is $f \in (M(G)_{ess})^\perp = \big(M(G)^*\big)_{ann}$; and $f(\mu_z) = \frac{1}{2} + \frac{1}{2} = 1$, so $\|f\| = 1$ and proposition 2.4 applied to the net above gives $\operatorname{sep}_{M(G)^*}(f) \geq 1$. Hence $\operatorname{sepd}(M(G)^*) = 1$. □

Replacing the proof of proposition `L1MetTopProjAndMetrFlatOfMeasAlg`$(iv) \Rightarrow (i)$. Delete the present argument, which opens with $M(G) \underset{L_1(G)\text{-}\mathbf{mod}_1}{\cong} L_1(G) \bigoplus_1 M_s(G)$ and concludes through propositions `MetTopFlatModCoProd` and `MetTopFlatAnnihModCharac`. Insert:

> $(iv) \Rightarrow (i)$ Suppose G is not discrete. By proposition 3.1 the $L_1(G)$-module $M(G)$ has asymptotic separation defect 1, and $L_1(G) \neq \{0\}$; so corollary 2.32 says that $M(G)$ is not metrically flat. Hence a metrically flat $M(G)$ forces G to be discrete.

Items (i)–(iii) of that proposition stand as printed. The same estimate bounds the topological constant from below, which the present text does not record.

Corollary 3.2. *Let G be a non-discrete locally compact group. Then every constant of topological flatness of the $L_1(G)$-module $M(G)$ is at least 2. The module is topologically flat by proposition* `MeasAlgIsL1TopFlat`*, so the least such constant lies in $[2, +\infty)$.*

Proof. By proposition 3.1 we have $\operatorname{asepd}(M(G)) = 1$, and $L_1(G) \neq \{0\}$; apply corollary 2.32. □

3.3 For subsection `SubSectionc0AndlInftyModules`, the metric half of proposition `c0AndlInftyModlIfty`(ii)

Replace the appeal to proposition `CStarAlgIsTopFlatOverItsIdeal` by the asymptotic separation defect. The paragraph becomes:

> (ii) Let Λ be infinite. We claim that $\operatorname{asepd}(\ell_\infty(\Lambda)) = 1$ over $c_0(\Lambda)$. The essential part is $\operatorname{cl}(c_0(\Lambda)\ell_\infty(\Lambda)) = c_0(\Lambda)$. For a finite $S \subset \Lambda$ put $v_S := \chi_{\Lambda\setminus S}$; these are unit vectors, they satisfy $\|av_S\|_\infty = \sup_{\lambda\notin S}|a(\lambda)| \to 0$ for every $a \in c_0(\Lambda)$ as S runs over the finite subsets of Λ directed by inclusion, so they form an annihilator net in $B_{\ell_\infty(\Lambda)}$; and $\operatorname{sep}_{\ell_\infty(\Lambda)}(v_S) = \operatorname{dist}(v_S, c_0(\Lambda)) = 1$, a norming direction being $f(x) = \lim_{\mathfrak{U}} x(\lambda_n)$ along a free ultrafilter $\mathfrak{U}$ on $\mathbb{N}$ and a sequence of distinct $\lambda_n \in \Lambda \setminus S$, which vanishes on $c_0(\Lambda)$, has norm at most one and takes the value 1 at v_S. Since $c_0(\Lambda) \neq \{0\}$, corollary 2.32 applies and gives that $\ell_\infty(\Lambda)$ is not a metrically flat $c_0(\Lambda)$-module. Both claims regarding projectivity follow from proposition `MetTopProjIsMetTopFlat`. If Λ is finite, then $c_0(\Lambda) = \ell_\infty(\Lambda)$ and the result follows from paragraph (i).

The same estimate gives, as prose after the proposition, that every constant of topological flatness of the $c_0(\Lambda)$-module $\ell_\infty(\Lambda)$ is at least 2 for infinite Λ — a quantitative refinement that the appeal to proposition `CStarAlgIsTopFlatOverItsIdeal` does not give. The topological half of (ii) is unaffected and keeps its present proof.

4 Merge manifest

4.1 Preamble additions required in `book/main.tex`

The following go beside the existing definitions of `\projtens`, `\convol`, `\projmodtens` and `\isom`. Nothing else in the preamble changes, and in particular `\sp` is left alone: [B] uses $\operatorname{sp}_A(a)$ for the spectrum of an algebra element throughout chapter 3, so the separation operator is called sep and not sp.

```
\DeclareMathOperator{\dist}{dist}
\DeclareMathOperator{\sep}{sep}
\DeclareMathOperator{\sepd}{sepd}
\DeclareMathOperator{\asep}{asep}
\DeclareMathOperator{\asepd}{asepd}
\DeclareMathOperator{\sng}{sng}
\newcommand{\cri}{c_{\mathrm{ri}}}
\newcommand{\cli}{c_{\mathrm{li}}}
\newcommand{\crai}{c_{\mathrm{rai}}}
\newcommand{\clai}{c_{\mathrm{lai}}}
```

The four constants may equally be spelled out in the text; the operators cannot, and sng is the only one of the six whose name was free in [B] already.

4.2 Chapter 1

- *Insertion*, subsection `SubSectionBanachSpaces`, in the paragraph carrying $E^*/F^\perp \underset{\mathbf{Ban}_1}{\cong} F^*$: the dist notation, the Hahn-Banach distance formula, Banach-Alaoglu, Goldstine and Mazur. Part I.1. Prose, no numbered statement.

- *Insertion*, subsection `SubSectionBanachAlgebrasAndTheirModules`, after the paragraph on approximate identities: the four constants $c_{\mathrm{ri}}, c_{\mathrm{li}}, c_{\mathrm{rai}}, c_{\mathrm{lai}}$, their conventions, the elementary relations, the open question, and the forward pointer to attainment. Part I.2.

- *Insertion*, same subsection: the definition of an annihilator net, with the disclaimer that no norm bound is part of it. Part I.3.

- *Insertion*, same subsection, in the paragraph that already records $(X^*)^{\perp S} = \mathrm{cl}_X(SX)^{\perp}$: the two dual identities spelled out at $S = A$. Part I.4. Note that [B] already has the first of them; only the second, and the remark that $(X^*)_{ess} \subset (X_{ann})^{\perp}$ can be proper, are new.

4.3 Chapter 2

- *Renumbering.* A new section 2.2, "The essential and the annihilator part", is inserted between the present 2.1 and the present 2.2. The present subsection 2.2.1, `SubSectionHomoligicallyTrivialAnnihil` becomes subsection 2.2.6 of the new section; the rest of the present section 2.2 becomes section 2.3, and the present section 2.3 becomes section 2.4. No label of [B] changes, and no statement of [B] is renumbered by hand: every citation in this note is by label.

- *Relocation, no change of content*, into subsection 2.3 at its head: proposition `MetTopRelProjModCompIdealPa` corollary `MetTopRelProjModPartCompl`, proposition `MetTopProjModEssQuotIsProjBanSp`.

- *Relocation, no change of content*, into subsection 2.4 at its head: proposition `MetTopRelInjModCompIdealAnn` corollary `MetTopRelInjModAnnihCompl`, proposition `MetTopInjModAnnihPartIsInjBanSp`.

- *Relocation, no change of content*, into subsection 2.5 at its head: proposition `MetTopRelFlatModCompIdealPa` corollary `MetTopRelFlatModPartCompl`, proposition `MetTopFlatModEssQuotIsFlatBanSp`.

- *Insertion*, subsections 2.1 to 2.5: everything else in Part II, in the order printed there.

- *Replacement*, proposition `MetTopProjModCCharac` and proposition `MetTopProjOfAnnihModCharac`, the algebra clause of $(i) \Rightarrow (ii)$: by corollary 2.18. See the caveat in subsection 2.6 and item (c) below.

- *Replacement*, proposition `MetTopInjModCCharac` and proposition `MetTopInjOfAnnihModCharac`, the algebra clause of $(i) \Rightarrow (ii)$: by corollary 2.23.

- *Replacement*, proposition `MetTopFlatAnnihModCharac`, the algebra clause of $(i) \Rightarrow (ii)$: by corollary 2.28.

- *Insertion*, subsection 2.6, after proposition `MetTopFlatAnnihModCharac`: corollary 2.33.

- *Labelling*, subsection 2.6: the proposition asserting that no Banach algebra makes all of its modules flat carries no label in [B] and cannot be cited. Proposed label: `NoAlgWithAllModFlat`.

4.4 Chapter 3

- *Deletion and replacement*, subsection `SectionPreliminariesOnHarmonicAnalysis`: the coproduct, the subalgebra and the annihilator-module assertions about $M_s(G)$, replaced by the text of Part III.1.

- *Insertion*, subsection `SubSectionL1GModules` at its head: proposition 3.1.

- *Replacement*, proposition `L1MetTopProjAndMetrFlatOfMeasAlg`, the proof of $(iv) \Rightarrow (i)$: by the three lines of Part III.2. Items (i)–(iii) stand as printed.

- *Insertion*, subsection `SubSectionL1GModules`, after proposition `MeasAlgIsL1TopFlat`: corollary 3.2.

- *Deletion*, proposition `MeasAlgIsL1TopFlat`: its opening paragraph, which proves $M(G)_{ess} = M_a(G)$ a second time, replaced by a citation of proposition 3.1.

- *Replacement*, proposition `c0AndlInftyModlIfty`(ii), the metric half: by the paragraph of Part III.3, which removes the appeal to proposition `CStarAlgIsTopFlatOverItsIdeal`. The topological half is unaffected.

4.5 Points to vet before merging

(a) *The $K = 0$ gap in lemma 2.2* is closed by disposing of $K = 0$ in a sentence, not by imposing $K \geq 1$. Imposing $K \geq 1$ would be consistent with [B]'s notion of a c-bounded approximate identity, but it would break the lemma's own consequence at $A = \{0\}$, where corollary 2.29 invokes it with $K = c_{\mathrm{lai}}(A) + \epsilon = \epsilon < 1$. Since $K \geq 1$ is automatic for $A \neq \{0\}$, the hypothesis would bite only on the zero algebra — the exact case the rest of the sweep is trying to include.

(b) *The zero algebra in subsection 2.2.* The hypothesis $A \neq \{0\}$ has been dropped from theorem 2.11 altogether, and all three clauses were checked at $A = \{0\}$: $(A_+)_{ess} = \{0\}$, $(A_+^*)_{ann} = A_+^* = \mathbb{C}\varepsilon$, $(A_+^*)_{ess} = \{0\}$, so $\mathrm{sepd}(A_+^*) = 1 = 1/(1 + c_{\mathrm{rai}}(\{0\}))$; the zero net is admissible in the lower bound of (ii) and attains the value in (iii), where $\delta = 1$ forces $m = 0$ and $p = 0$; and (i) reads $\mathrm{sep}_{X^*}(f) \geq \|f\|$, which holds with equality because $X_{ess} = \{0\}$. The hypothesis is kept exactly where it bites: lemma 2.12, which needs an element of norm one, and hence corollary 2.13, whose first clause fails at $A = \{0\}$. The asymmetry with lemma 2.10, which never needed the hypothesis, is thereby removed, and the zero-algebra paragraph formerly buried in the proof of corollary 2.27 disappears, along with the case splits it forced on corollaries 2.28 and 2.29.

(c) *Attainment of $c_{\mathrm{ri}}(A)$.* The separation estimate proves $c_{\mathrm{ri}}(A) \leq C - 1$, whereas propositions `MetTopProjModCCharac` and `MetTopProjOfAnnihModCharac` as printed assert a right identity of norm at most $C - 1$. The infimum defining $c_{\mathrm{ri}}(A)$ need not be attained: the right identities of A form the closed coset $u + \{z \in A : Az = \{0\}\}$, and closed subspaces of a Banach space are not in general proximinal. This is a real difference from the injective and flat clauses, where theorem 2.11(iii) does give attainment. Recommendation: restate the two projective clauses as $c_{\mathrm{ri}}(A) \leq C - 1$. If the attained form is to be kept, the appeal to proposition `MetTopProjCycModCharac` must be kept with it; that is legal, since it lies in section 2.1.

(d) *Theorem 2.11(iii) has been retained.* The instruction was to check whether the attainment of $c_{\mathrm{rai}}(A)$ is classical — Doran and Wichmann, [`AppIdAndFactorInBanAlg`] — and, if so, to cite it and delete (iii). That could not be verified against the source, and deleting a short self-contained proof on the strength of an unverified citation seemed the worse risk. The clause is used in corollaries 2.23, 2.28 and 2.33 and in corollary 2.13, so if the reference is confirmed it is those four citations that change, not the argument.

(e) *One label is split.* The former `InjAnnihModApprIdentOptimal` had two clauses. Clause (i) is corollary 2.23 and stays in subsection 2.4; clause (ii) keeps the old label and moves to subsection 2.6. The move is forced: the proof of (ii) quotes propositions `MetTopInjOfAnnihModCharac` and `MetTopFlatAnnihModCharac` in the direction $(ii) \Rightarrow (i)$, and those live in subsection 2.2.6. Leaving it in 2.2.4 would be a forward reference of exactly the kind the reorganization is meant to remove.

(f) *One statement is shortened.* Corollary 2.18 is stated as a necessary condition only. Its converse, in the note, was a citation of proposition `MetTopProjOfAnnihModCharac`, which now sits later; and since the corollary dissolves into that very proposition, nothing is lost.

(g) *Proof-internal cross-references removed, beyond the four the instruction listed.* Three more were found and are handled the same way. The identity $X_{ann} = \bigoplus_p \{(X_\lambda)_{ann}\}$ is now part of the statement of proposition 2.9, since corollary 2.24(ii) used it. The identities $(A_+^*)_{ann} = \mathbb{C}\varepsilon$, $\|\varepsilon\| = 1$ and $\mathrm{sepd}(A_+^*) = \mathrm{sep}_{A_+^*}(\varepsilon)$ are now part of the statement of lemma 2.10, since proposition 2.21, corollary 2.27 and proposition 2.14 all quoted them from inside the proof of theorem 2.11(ii). And proposition 2.15, which quoted the essential part of the free module from inside the proof of proposition 2.14, now derives it from proposition 2.9(iv) directly.

(h) *The reading of proposition `MetTopFlatCycModCharac`* recorded at the end of subsection 2.2 is stated for the unitization A_+. It is there, and only there, that $\|e_{A_+} - e_\nu\| = 1 + \|e_\nu\|$ for $e_\nu \in I \subset A$; over $A_\times$ the quantity is not $1/\,\mathrm{sepd}(I_+^*)$.

(i) *Length.* No proof in Part II exceeds the longest proofs of chapter 2 of [B], namely those of lemma `GoodIdealMetTopProjIsUnital` and proposition `MetTopFlatCycModCharac`. The step-numbered format is gone everywhere, and the verifications [B] habitually compresses — three-epsilon arguments, checks that a constructed map is a morphism, directing a net over pairs (F, ϵ), and the 1-Lipschitz and positively homogeneous character of sep_X — are compressed here as well, the last being stated once after definition 2.1 and used silently thereafter.